HESAI PANDARXT-16 USER MANUAL

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PandarXT-16

16-Channel Medium-Range Mechanical LiDAR User Manual

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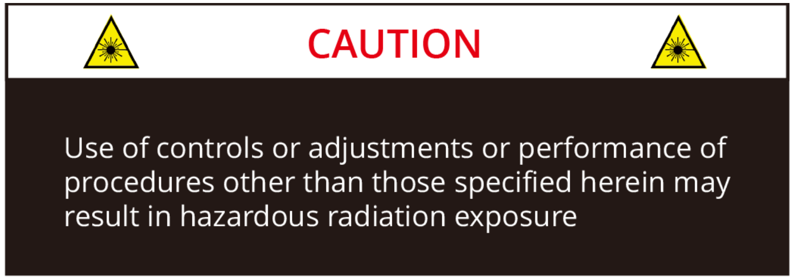
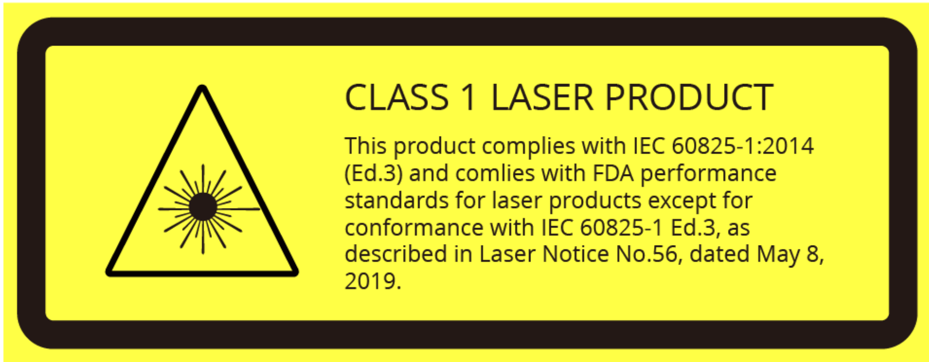
Safety Notice

PLEASE READ AND FOLLOW ALL INSTRUCTIONS CAREFULLY AND CONSULT ALL RELEVANT NATIONAL AND INTERNATIONAL SAFETY REGULATIONS FOR YOUR APPLICATION.

■ Caution

To avoid violating the warranty and to minimize the chances of getting electrically shocked, please do not disassemble the product. The product must not be tampered with and must not be changed in any way. There are no user-serviceable parts inside the product. For repairs and maintenance inquiries, please contact an authorized Hesai Technology service provider.





■ Laser Safety Notice - Laser Class 1

This device satisfies the requirements of

- IEC 60825-1:2014
- 21 CFR 1040.10 and 1040.11 except for deviations (IEC 60825-1 Ed.3) pursuant to Laser Notice No.56, dated May 8, 2019

NEVER LOOK INTO THE TRANSMITTING LASER THROUGH A MAGNIFYING DEVICE (MICROSCOPE, EYE LOUPE, MAGNIFYING GLASS, ETC.)

Safety Precautions

In all circumstances, if you suspect that the product malfunctions or is damaged, stop using it immediately to avoid potential hazards and injuries. Contact an authorized Hesai Technology service provider for more information on product disposal.

Handling

This product contains metal, glass, plastic, as well as sensitive electronic components. Improper handling such as dropping, burning, piercing, and squeezing may cause damage to the product.

In case the product is dropped, STOP using the product immediately and contact Hesai technical support.

Enclosure

This product contains high-speed rotating parts. To avoid potential injuries, DO NOT operate the product if the enclosure is loose or damaged. To ensure optimal performance, do not touch the product's enclosure with bare hands. If the enclosure is already stained, please refer to the cleaning method in the Sensor Maintenance chapter of user manuals.

Eye Safety

Although the product meets Class 1 eye safety standards, DO NOT look into the transmitting laser through a magnifying product (microscope, eye loupe, magnifying glass, etc.). For maximum self-protection, avoid looking directly at the product when it is in operation.

Repair

DO NOT open and repair the product without direct guidance from Hesai Technology. Disassembling the product may cause degraded performance, failure in water resistance, or potential injuries to the operator.

Power Supply

Use only the cables and power adapters provided by Hesai Technology. Only the power adapters that meet the product's power requirements and applicable safety standards can be used. Using damaged cables, adapters or supplying power in a humid environment can result in fire, electric shock, personal injuries, product damage, or property loss.

Hot Surface

During or after a period of operation, DO NOT touch the product's enclosure with your skin. Such direct contact with the hot surface can result in discomfort or even burns. If you incorporate this LiDAR product into your product(s), you should also communicate the hot surface risks to the intended users of your product(s).

Vibration

Strong vibration may cause damage to the product and should be avoided. If you need the mechanical vibration and shock limits of this product, please contact Hesai technical support.

Radio Frequency Interference

Please observe the signs and notices on the product that prohibit or restrict the use of electronic devices. Although the product is designed, tested, and manufactured to comply with the regulations on RF radiation, the radiation from the product may still influence electronic devices.

Medical Device Interference

Some components in the product can emit electromagnetic fields, which may interfere with medical devices such as cochlear implants, heart pacemakers, and defibrillators. Consult your physician and medical device manufacturers for specific information regarding your medical device(s) and whether you need to keep a safe distance from the product. If you suspect that the product is interfering with your medical device, stop using the product immediately.

Explosive Atmosphere and Other Air Conditions

Do not use the product in any area where potentially explosive atmospheres are present, such as high concentrations of flammable chemicals, vapors, or particulates (including particles, dust, and metal powder) in the air. Exposing the product to high concentrations of industrial chemicals, including liquefied gases that are easily vaporized (such as helium), can damage or weaken the product's function. Please observe all the signs and instructions on the product.

Light Interference

Some precision optical instruments may be interfered by the laser light emitted from the product.

1 Introduction

This manual describes the specifications, installation, and data output format of PandarXT-16.

This manual is under constant revision. To obtain the latest version, please visit the Download page of Hesai's official website, or contact Hesai technical support.

1.1 Operating Principle

Distance Measurement: Time of Flight (ToF)

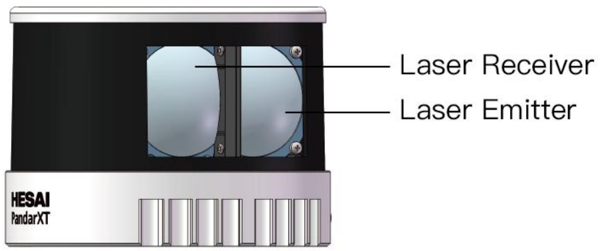
1) A laser diode emits a beam of ultrashort laser pulses onto the target object.
2) The laser pulses are diffusely reflected after hitting the target object. The returning beam is detected by an optical sensor.
3) Distance to the object can be accurately measured by calculating the time between laser emission and receipt.

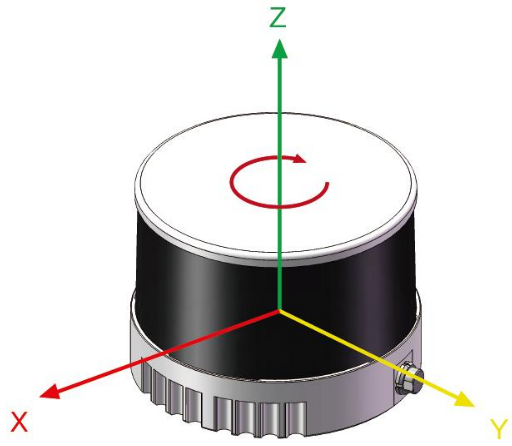
$d = \frac{ct}{2}$	d: distance c: speed of light t: travel time of the laser beam

Figure 1.1 Distance Measurement Using Time of Flight

1.2 LiDAR Structure

Laser emitters and receivers are attached to a motor that rotates horizontally.





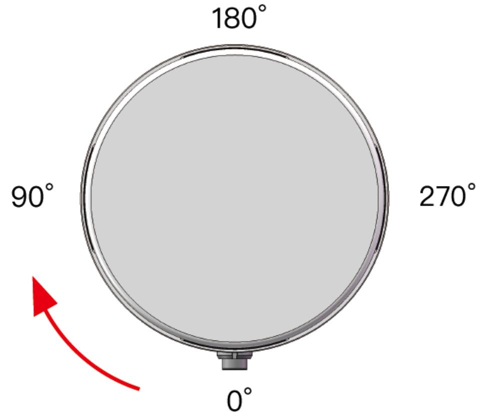


Figure 1.2 Partial Cross-Sectional Diagram Figure 1.3 Coordinate System (Isometric View) Figure 1.4 Rotation Direction (Top View)

The LiDAR's coordinate system is illustrated in Figure 1.3. Z-axis is the axis of rotation.

The origin is shown as a red dot in Figure 1.6 on the next page. All measurements are relative to the origin.

When all channels pass the zero-degree position in Figure 1.4, the azimuth data in the corresponding UDP data block will be 0°.

1.3 Channel Distribution

The vertical resolution is 2° across the FOV, as shown in Figure 1.5.

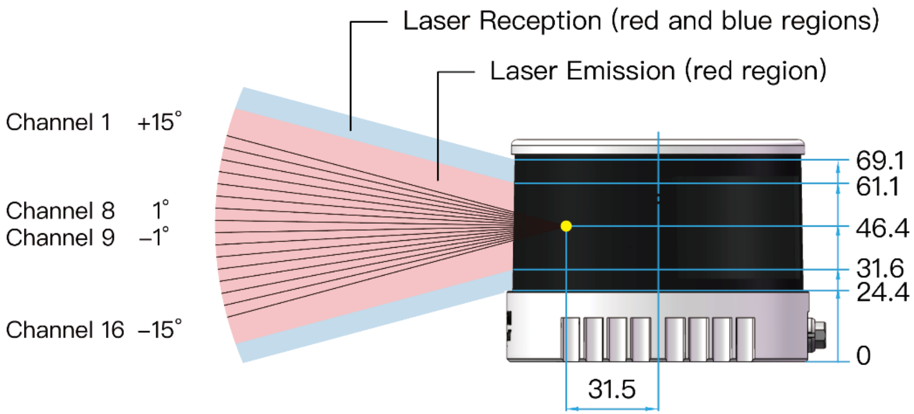


Figure 1.5 Channel Vertical Distribution

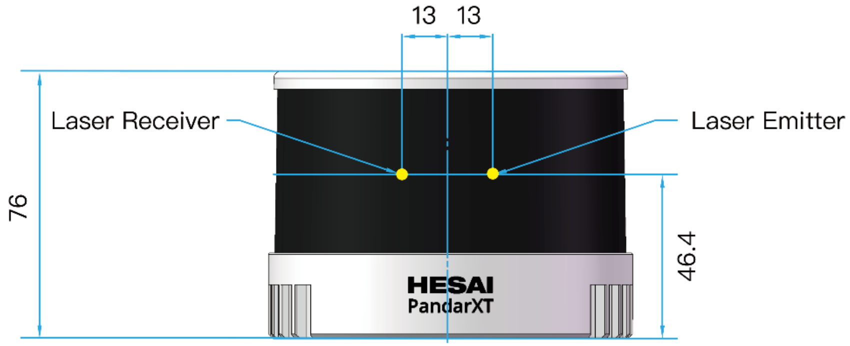


Figure 1.6 Laser Firing Position (Unit: mm)

Each channel has an intrinsic vertical angle offset.
The offsetted angles s are recorded in this LiDAR unit's calibration file, which is provided when shipping the unit.

In case you need to obtain the file again:

- Send this TCP command *PTC_COMMAND_GET_LIDAR_CALIBRATION*, as described in Hesai TCP API Protocol (Chapter 6).
- Or contact a sales representative or technical support engineer from Hesai.

1.4 Specifications

SENSOR	
Scanning Method	Mechanical Rotation
Channel	16
Instrument Range	0.05 to 120 m
Range Capability	80 m @10% reflectivity (Channels 5-12)
	50 m @10% (Channels 1-4，13-16)
Range Accuracy	±1 cm (typical)
Range Precision	0.5 cm (typical)
FOV (Horizontal)	360°
Resolution (Horizontal)	0.09° (5 Hz)
	0.18° (10 Hz)
	0.36° (20 Hz)
FOV (Vertical)	30° (-15° to +15°)
Resolution (Vertical)	2°
Frame Rate	5 Hz, 10 Hz, 20 Hz
Returns	Single Return (Last, Strongest)
	Dual Return (Last and Strongest)
CERTIFICATIONS	
	Class 1 Laser Product
	CE, FCC, FDA, IC, RCM, EAC, KCC

MECHANICAL/ELECTRICAL/OPERATIONAL	
Wavelength	905 nm
Laser Class	Class 1 Eye Safe
Ingress Protection	IP6K7
Dimensions	Height: 76.0 mm
	Top/Bottom Diameter: 100.0 / 103.0 mm
Operating Voltage	DC 9 to 36 V
Power Consumption	9 W (typical)
Operating Temperature	-20°C to 65°C
Weight	0.8 kg
DATA I/O	
Data Transmission	UDP/IP Ethernet (100 Mbps)
Data Outputs	Distance, Azimuth Angle, Intensity
Data Points Generated	Single Return: 320,000 points/sec
	Dual Return: 640,000 points/sec
Clock Source	GPS / PTP
PTP Clock Accuracy	⩽1 μs
PTP Clock Drift	⩽1 μs/s

NOTE Specifications are subject to change. Please refer to the latest version.

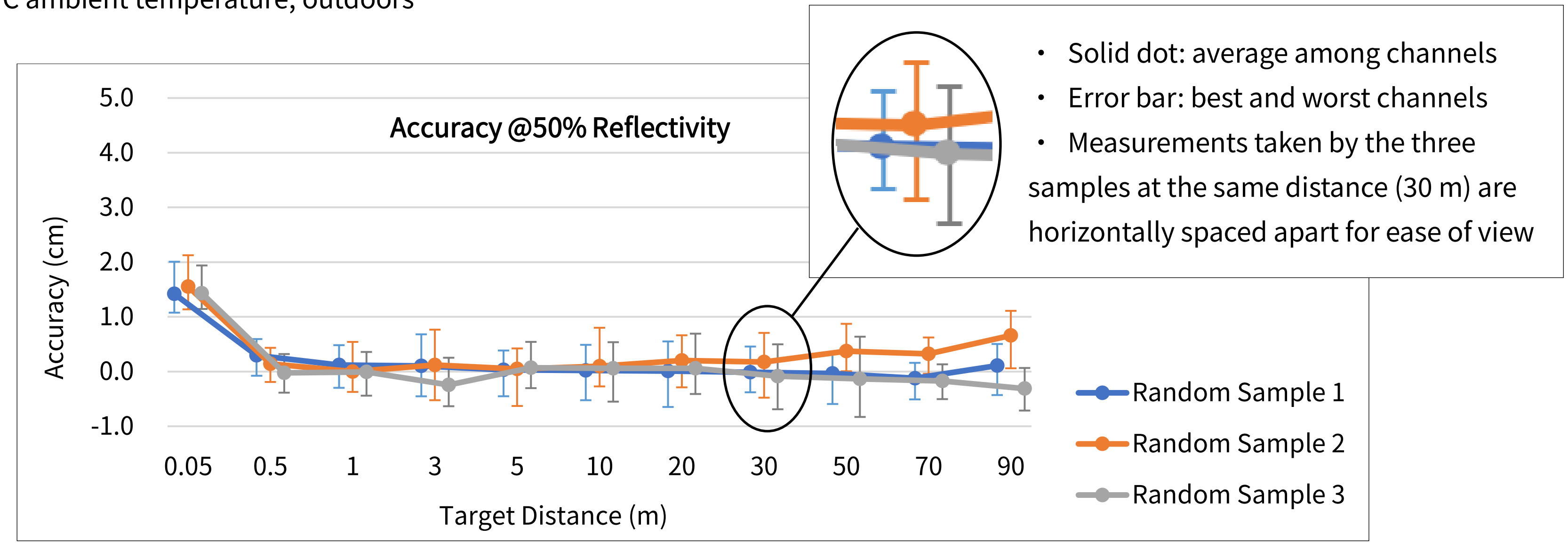
NOTE Range Capability: 100 klux ambient intensity, PoD (probability of detection) > 90%, FAR (false alarm rate) < 10E-5.

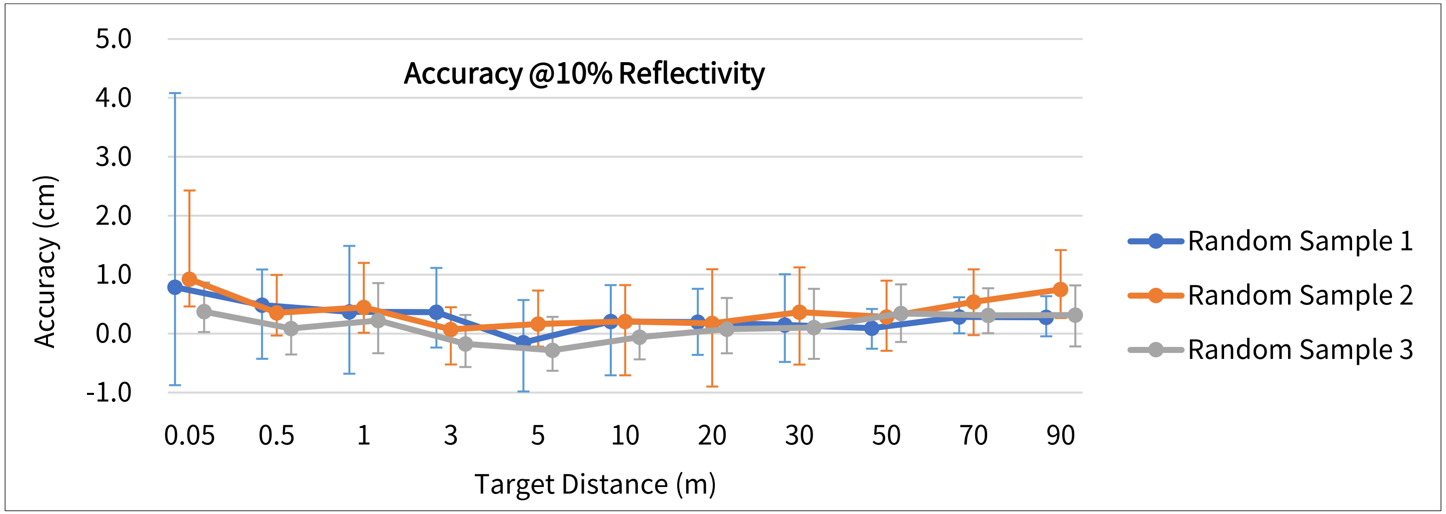
NOTE Range accuracy and precision: may vary with range, temperature, and target reflectivity. The typical values are the average among channels, measured outdoors within 0.5 ~ 70 m, under 30°C ambient temperature, and with a target reflectivity of 50%.

Range Accuracy

Definition: the average of the differences between multiple measurements and the target's true distance, measured by a single channel

Conditions: 30°C ambient temperature, outdoors

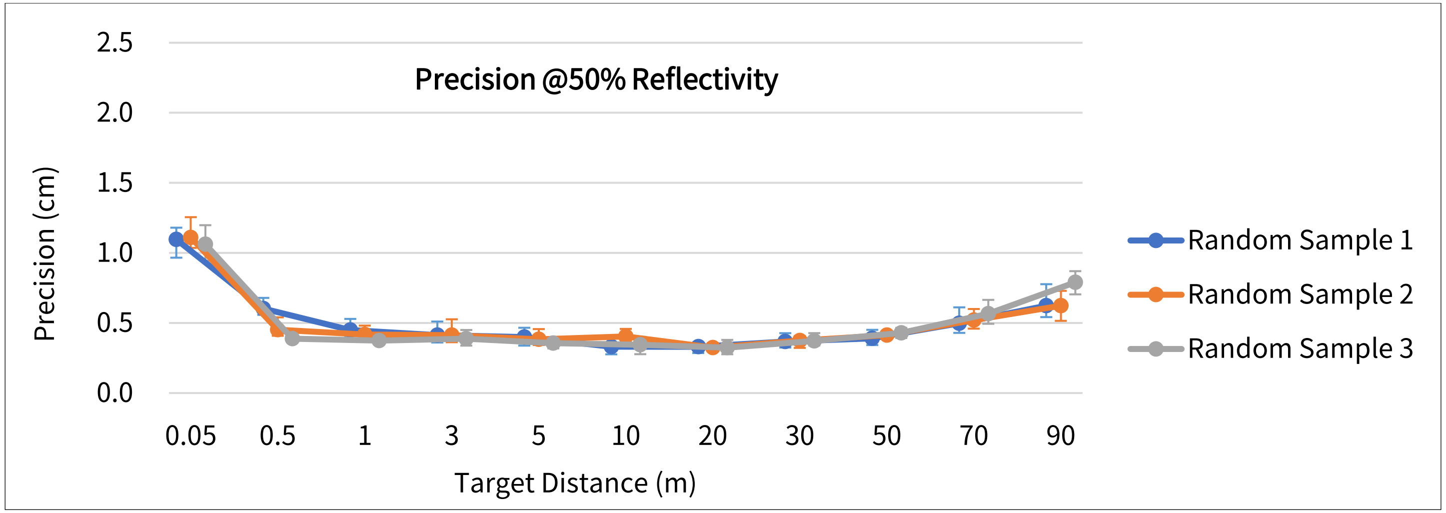


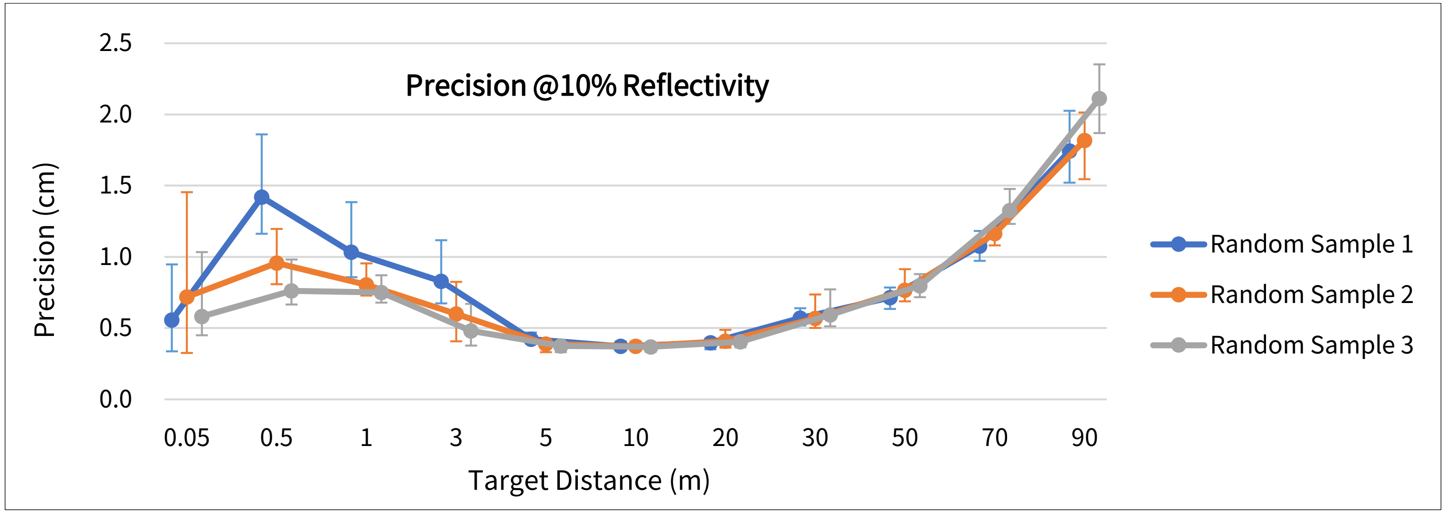


■ Range Precision

Definition: the standard deviation among multiple measurements, measured by a single channel

Conditions: 30°C ambient temperature, outdoors





2 Setup

2.1 Mechanical Installation

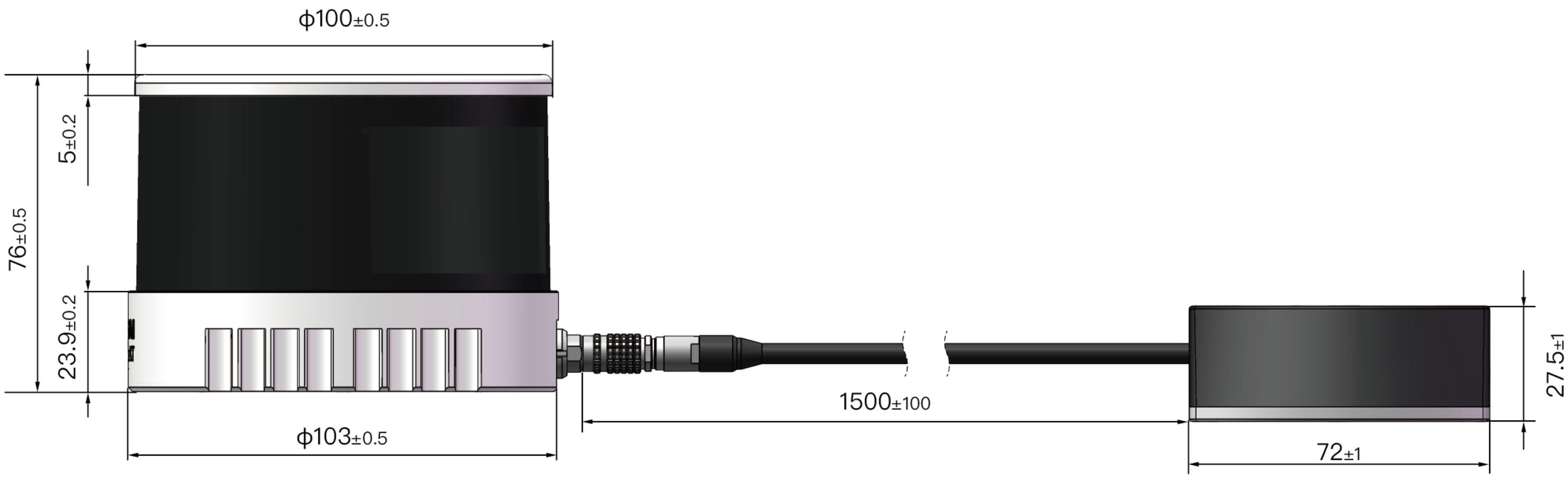


Figure 2.1 Front View (Unit: mm)

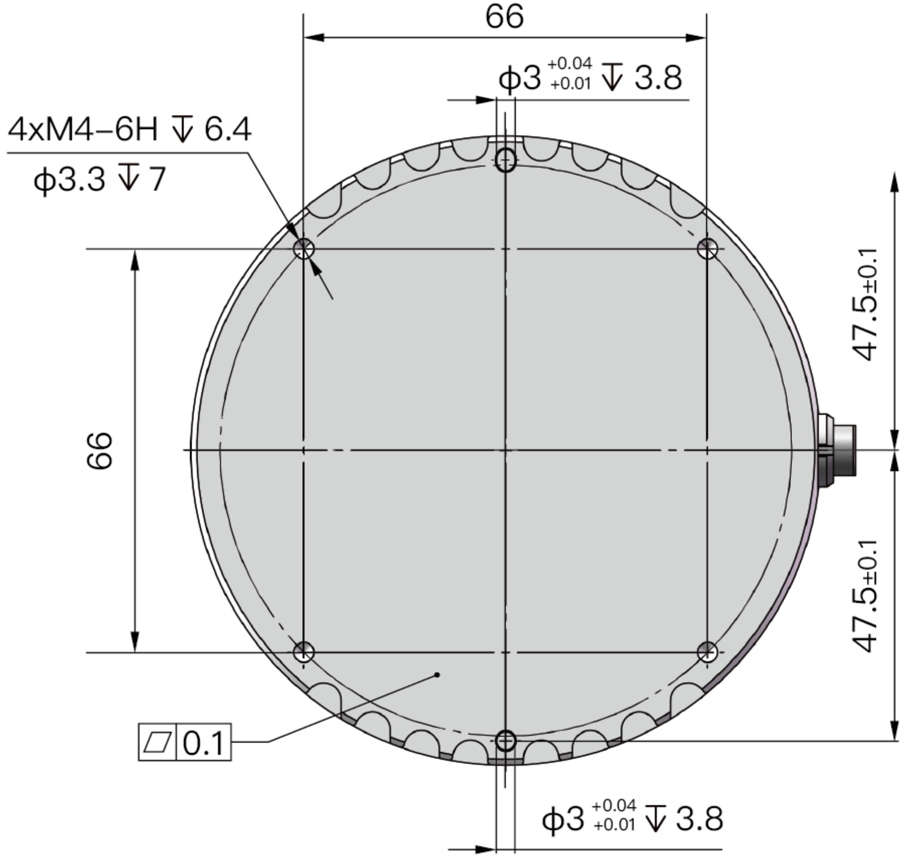


Figure 2.2 Bottom View (Unit: mm)

■ **Recommended Installation**

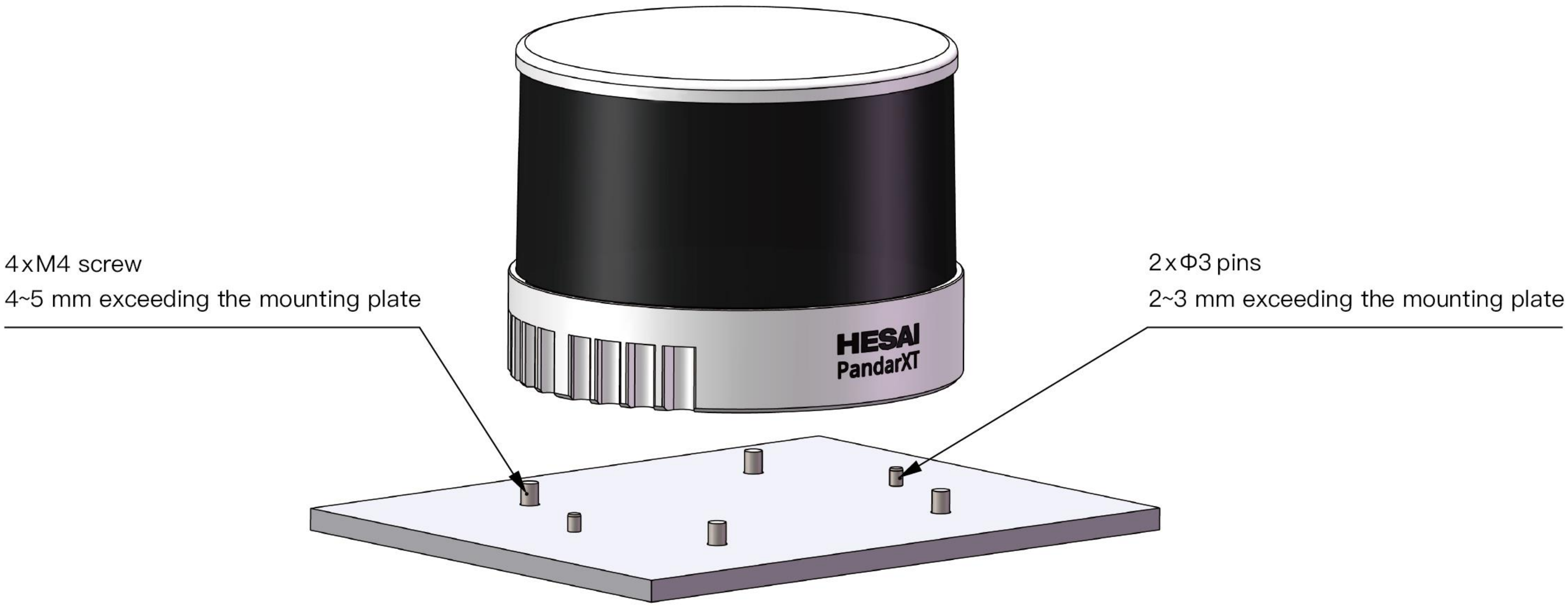


Figure 2.3 Recommended Installation

2.2 Interfaces

Lemo part number: EEG.0T.309.CLN (female socket, on the LiDAR)

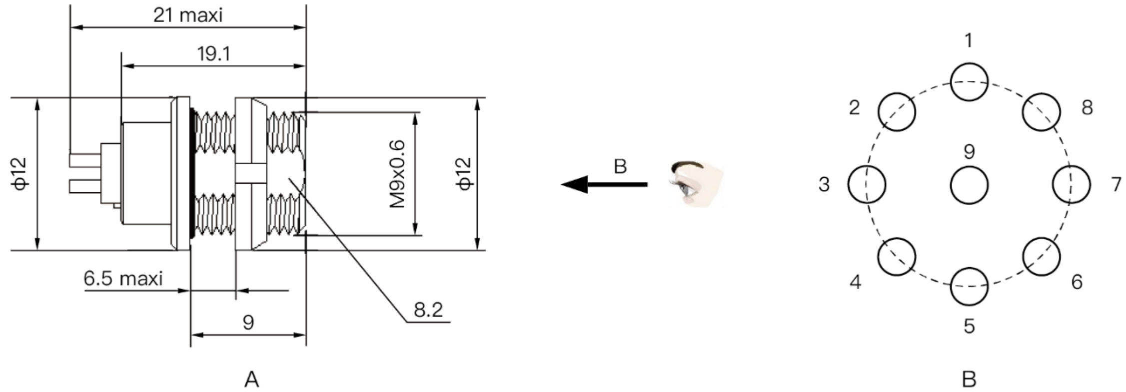


Figure 2.4 Lemo Connector (Female Socket)

Pin #	Signal	Color	Voltage
1	GPS PPS	BLACK	TTL 3.3/5 V
2	GPS DATA	PURPLE	-13 to +13 V
3	GND	BROWN	0 V
4	VIN	WHITE	9 to 36 V
5	Ethernet TX+	YELLOW	-1 to 1 V

Pin #	Signal	Color	Voltage
6	Ethernet TX-	GREEN	-1 to 1 V
7	Ethernet RX+	PINK	-1 to 1 V
8	Ethernet RX-	GRAY	-1 to 1 V
9	-	-	-

NOTE For the GPS PPS signal, pulse width is recommended to be over 1 ms, and the cycle is 1 s (rising edge to rising edge)

NOTE Before connecting or disconnecting an external GPS signal (either using the cable's GPS wire or via the connection box's GPS port), make sure the LiDAR is powered off. If the LiDAR has to stay powered on, make sure to:

- ground yourself in advance
- avoid touching the GPS wire or GPS port with bare hands

Connector Use

Connection	Disconnection
• Turn off the power source • Make sure red dot on the cable's plug faces upward • Push the plug straight into the LiDAR's socket	• Turn off the power source • Hold the plug's shell and pull the plug from the socket

NOTE

- DO NOT attempt to force open a connection by pulling on the cables or by twisting the connectors in any way. Doing so can loosen the connectors' shells, or even damage the contacts.
- In case a connector's shell is accidentally pulled off, stop using the connector and contact Hesai technical support.
- DO NOT attempt to assemble the connector's shell and cable collet; DO NOT connect a connector without its shell. Doing so may damage the LiDAR's circuits.

Before connection: make sure the red dot faces upward

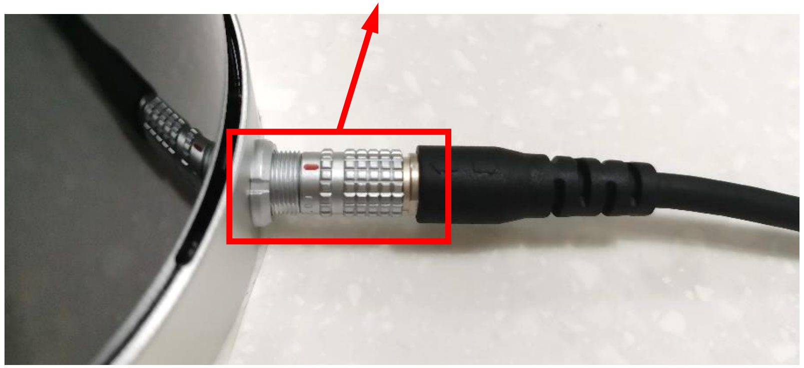


Figure 2.5 Lemo Connection

Cables

OD (outside diameter) = 5.0±0.2 mm

Minimum bend radius: 10 * OD

NOTE To avoid damaging the cable, do not bend the cable at the cable gland.

2.3 Connection Box (Optional)

Users may connect the LiDAR directly or using the connection box.
The connection box has a power port, a GPS port, and a standard Ethernet port.

Lemo part number: FGG.0T.309.CLAC50Z (male plug, on the connection box)

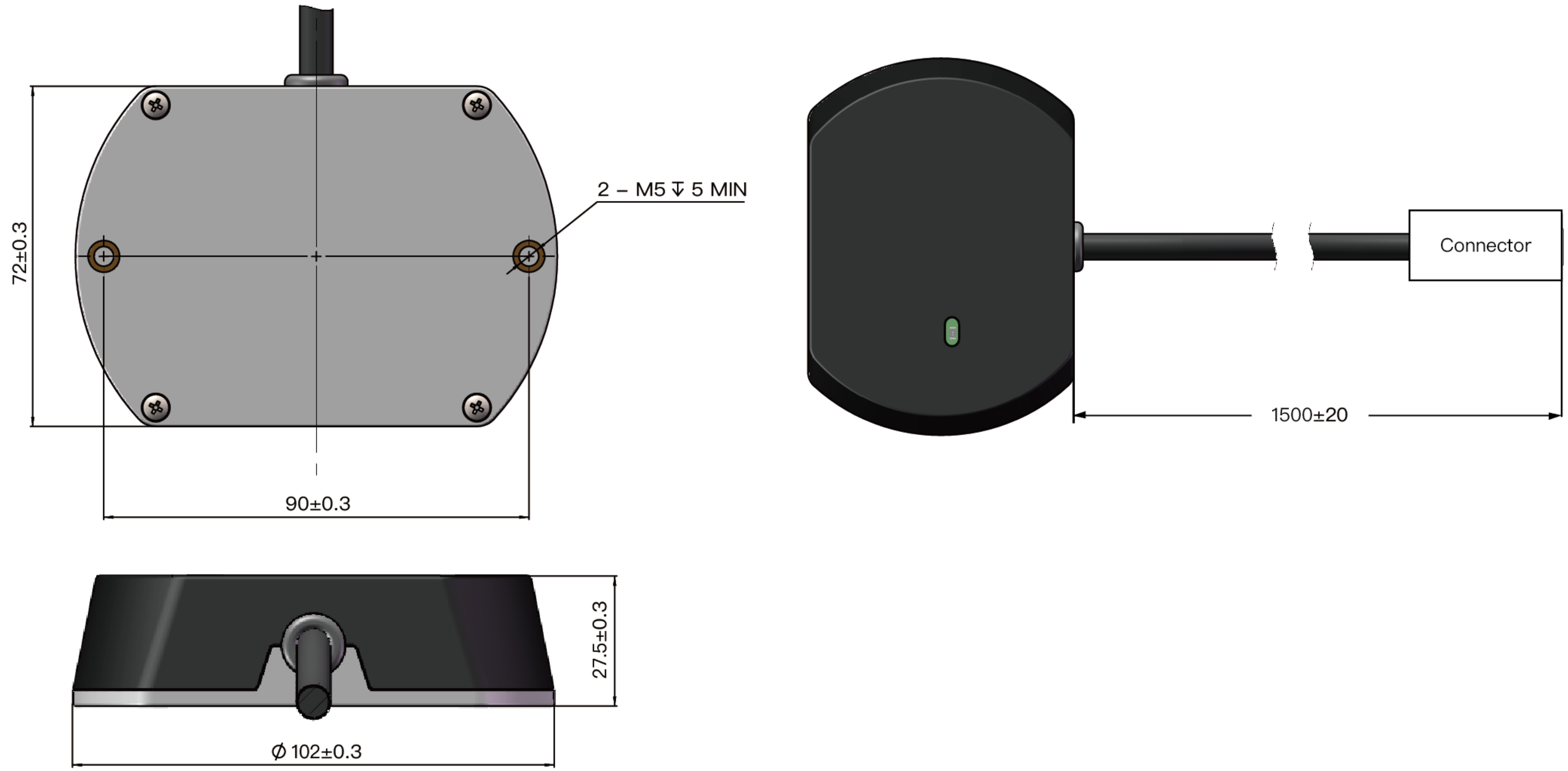


Figure 2.6 Connection Box (Unit: mm)

2.3.1 Connection Box Interfaces

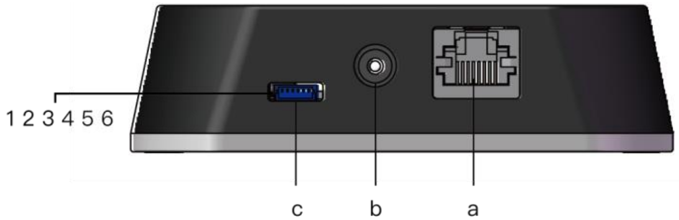


Figure 2.7 Connection Box (Front)

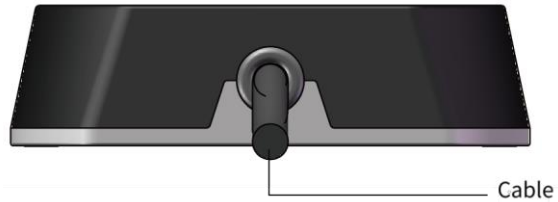


Figure 2.8 Connection Box (Back)

Port #	Port Name	Description
a	Standard Ethernet Port	RJ45, 100 Mbps Ethernet
b	Power Port	Connects to a DC power adapter External power supply: 9 V to 36 V, at least 30 W
c	GPS Port	Connector part number: JST SM06B-SRSS-TB Recommended connector for the external GPS module: JST SHR-06V-S-B Voltage standard: RS232 Baud rate: 9600 bps

The GPS port pin numbers are 1 to 6 from left to right, defined as follows:

Pin #	Direction	Pin Description	Requirements
1	Input	PPS (pulse-per-second) signal for synchronization	TTL level 3.3 V/5 V Recommended pulse width: ⩾1 ms Cycle: 1 s (from rising edge to rising edge)
2	Output	Power for the external GPS module	5 V
3	Output	Ground for the external GPS module	-
4	Input	Receiving serial data from the external GPS module	RS232 level
5	Output	Ground for the external GPS module	-
6	-	Reserved	-

2.3.2 Connection

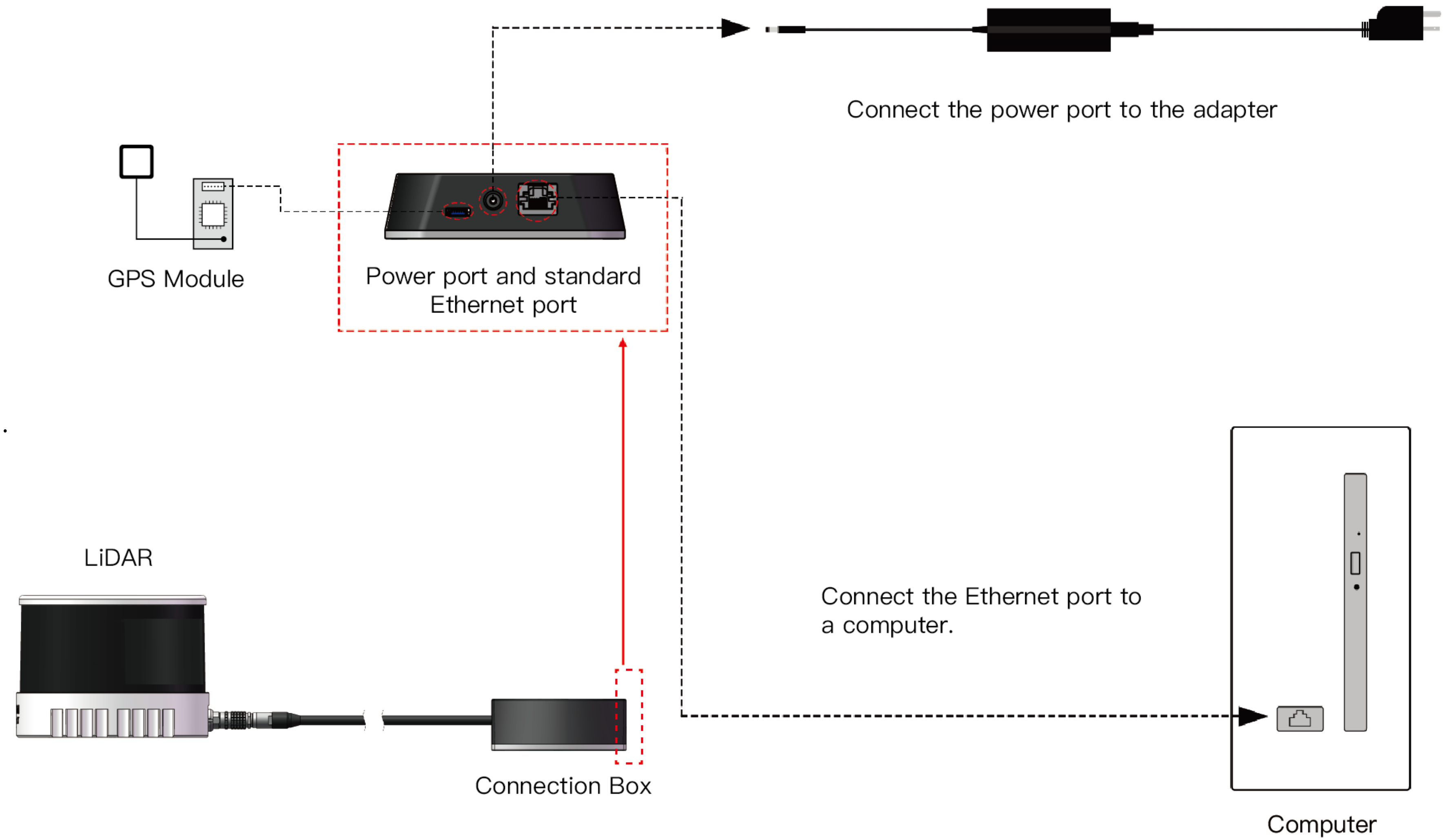


Figure 2.9 Connection Box - Connection

NOTE Refer to Appendix III when PTP protocol is used.

2.4 Get Ready to Use

Before operating the LiDAR, strip away the protective cover outside the enclosure.

The LiDAR does not have a power switch. It starts operating once connected to power and the Ethernet.

To receive data on your PC, set the PC's IP address to 192.168.1.100 and subnet mask to 255.255.255.0

For Ubuntu:	For Windows:
Input this ifconfig command in the terminal: ~$ sudo ifconfig enp0s20f0u2 192.168.1.100 (replace enp0s20f0u2 with the local Ethernet port name)	Open the Network Sharing Center, click on "Ethernet" In the "Ethernet Status" box, click on "Properties" Double-click on "Internet Protocol Version 4 (TCP/IPv4)" Configure the IP address to 192.168.1.100 and subnet mask to 255.255.255.0

To record and display point cloud data, see Chapter 5 (PandarView)

To set parameters, check device info, or upgrade firmware/software, see Chapter 4 (Web Control)

To obtain the SDKs (Software Development Kits) for your product model,
- please find the download link at: www.hesaitech.com/en/download (Product Documentation → select product model)
- or visit Hesai's official GitHub page: https://github.com/HesaiTechnology

3 Data Structure

The LiDAR outputs Point Cloud Data Packets and GPS Data Packets using 100 Mbps Ethernet UDP/IP.
Each data packet consists of an Ethernet header and UDP data.

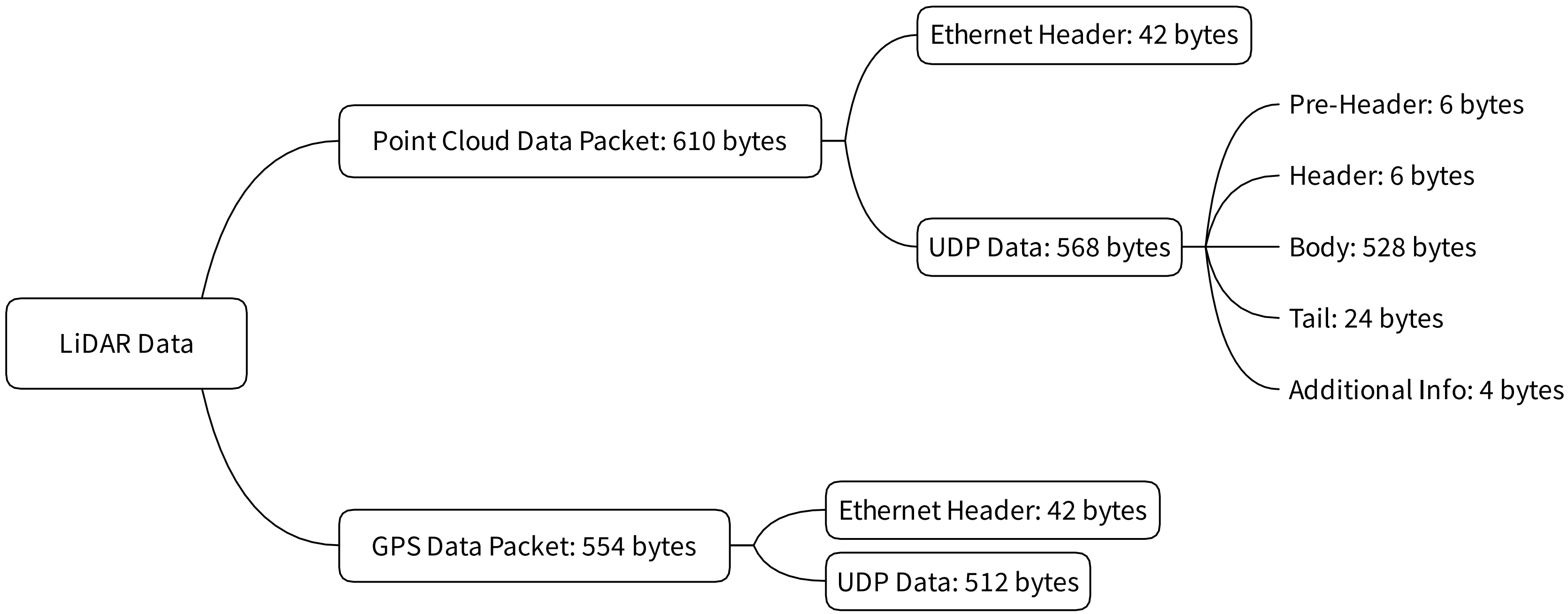


Figure 3.1 Data Structure

3.1 Point Cloud Data Packet

3.1.1 Ethernet Header

Each LiDAR has a unique MAC address. The source IP is 192.168.1.201 by default, and the destination IP is 255.255.255.255 (broadcast).

Point Cloud Ethernet Header: 42 bytes		
Field	Bytes	Description
Ethernet II MAC	12 bytes	Destination: broadcast (0xFF: 0xFF: 0xFF: 0xFF: 0xFF: 0xFF) Source: (xx:xx:xx:xx:xx:xx)
Ethernet Data Packet Type	2 bytes	0x08, 0x00
Internet Protocol	20 bytes	Shown in Figure 3.2
UDP Port Number	4 bytes	UDP source port (0x2710, representing 10000) Destination port (0x0940, representing 2368)
UDP Length	2 bytes	0x0240, representing 576 bytes (8 bytes more than the size of the Point Cloud UDP Data, shown in Figure 3.1)
UDP Checksum	2 bytes	-

```
Internet Protocol Version 4, Src: 192.168.1.201, Dst: 255.255.255.255
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 1108
    Identification: 0x7f1d (32541)
    Flags: 0x4000, Don't fragment
    Time to live: 64
    Protocol: UDP (17)
    Header checksum: 0xf50a [correct]
    [Header checksum status: Good]
    [Calculated Checksum: 0xf50a]
    Source: 192.168.1.201
    Destination: 255.255.255.255
```

Figure 3.2 Point Cloud Ethernet Header - Internet Protocol

3.1.2 UDP Data

All the multi-byte values are unsigned and in little endian format.

■ **Pre-Header**

Pre-Header: 6 bytes		
Field	Bytes	Description
0xEE	1	SOP (start of packet)
0xFF	1	SOP (start of packet)
Protocol Version Major	1	Major version number of the protocol: to distinguish between product models 0x06 for PandarXT
Protocol Version Minor	1	Minor version number of the protocol: for each product model, to indicate the current protocol version Currently 0x01 for PandarXT
Reserved	2	-

Header

Header: 6 bytes		
Field	Bytes	Description
Laser Num	1	0x10 (16 channels)
Block Num	1	0x08 (8 blocks per packet)
First Block Return	1	The first block in this data packet 0x00 - Single Return 0x01 - Last Return in Dual Return mode
Dis Unit	1	0x04 (4 mm)
Return Number	1	Number of returns that each channel generates 0x01 - one return 0x02 - two returns
UDP Seq	1	[7:1] is reserved Least significant bit [0] shows whether this packet includes a UDP sequence number field 0 - UDP sequence OFF 1 - UDP sequence ON **NOTE** Always 0x01 for the PandarXT series

■ **Body**

Body: 528 bytes (8 blocks)				
Block 1	**Block 2**	**Block 3**	**...**	**Block 8**
Azimuth 1	Azimuth 2	Azimuth 3	...	Azimuth 8
Channel 1	Channel 1	Channel 1	...	Channel 1
Channel 2	Channel 2	Channel 2	...	Channel 2
...	...	...	...	...
Channel 16	Channel 16	Channel 16	...	Channel 16

Under the Dual Return mode, the measurements from each round of firing are stored in two adjacent blocks:

- The odd number block is the last return, and the even number block is the strongest return
- If the last and strongest returns coincide, the second strongest return will be placed in the even number block
- The Azimuth changes every two blocks

Block size = size of Azimuth + 16 * size of Channel X

<table>
<tr><th colspan="4">Each Block in the Body: 66 bytes</th></tr>
<tr><td>Field</td><td>Bytes</td><td colspan="2">Description</td></tr>
<tr><td>Azimuth</td><td>2</td><td colspan="2">Current reference angle of the rotor, in little endian format (lower byte first)
Azimuth Angle = Azimuth / 100°</td></tr>
<tr><td rowspan="3">Channel X</td><td rowspan="3">4</td><td>2-byte Distance</td><td>Distance Value = Distance * 4
In little endian format (lower byte first)</td></tr>
<tr><td>1-byte Reflectivity</td><td>Reflectivity Value = Reflectivity * 1%
Range: 0 to 255</td></tr>
<tr><td>Reserved</td><td>-</td></tr>
</table>

Tail

Tail: 24 bytes		
Field	Bytes	Description
Reserved	9	-
High Temperature Shutdown Flag	1	0x01 for high temperature; 0x00 for normal operation ・ When high temperature is detected, the shutdown flag will be set to 0x01, and the system will shut down after 60 s. The flag remains 0x01 during the 60 s and the shutdown period ・ When the system is no longer in high temperature status, the shutdown flag will be reset to 0x00 and the system will automatically return to normal operation
Return Mode	1	0x37 for Strongest Return mode, 0x38 for Last Return mode, and 0x39 for Dual Return mode
Motor Speed	2	speed_2_bytes [15:0] = speed (RPM)
Date & Time	6	Year (current year minus 1900), month, date, hour, minute, second Binary, 1 byte each
Timestamp	4	The "μs time" part of the absolute time of this data packet (defined in Appendix II) Unit: μs Range: 0 to 1000000 μs (1 s)
Factory Information	1	0x42

Additional Info

Additional Info: 4 bytes		
Field	Bytes	Description
UDP Sequence	4	Sequence number of this UDP packet 0 to 0xFF FF FF FF in little endian format

3.1.3 Point Cloud Data Analysis

The analysis of point cloud UDP data consists of three steps.

■ **Analyze the vertical angle, horizontal angle, and distance of a data point**

Take PandarXT-16's Channel 5 in Block 2 as an example:

1) Vertical angle of Channel 5 is 7°, according to Appendix I (Channel Distribution)
- 0° represents the horizontal direction
- Define upward as positive
- The Channel # from the uppermost counts from 1

NOTE The accurate vertical angle is recorded in this LiDAR's unit's calibration file, see Section 1.3 (Channel Distribution).

2) Horizontal angle = current reference angle of the rotor + horizontal angle offset + firing time offset
- Current reference angle of the rotor: Azimuth field of Block 2
- Horizontal angle offset: 0° for Channel 5, according to Appendix I (Channel Distribution)
- Firing time offset = Laser Firing Time of Channel 5 (see Appendix II) * Spin Rate of the Motor (see Section 4.1 Web Control - Home)
- Define clockwise in the top view as the horizontal angles' positive direction

3) Actual distance in real world millimeters = distance measurement * Distance Unit (4 mm)
Distance measurement is the Distance field of Channel 5 in Block 2

■ **Draw the data point in a polar or rectangular coordinate system**

■ **Obtain the real-time point cloud data by analyzing and drawing every data point in each frame**

3.2 GPS Data Packet

GPS Data Packets are triggered every second. All the multi-byte values are unsigned and in little endian format.

Before NMEA messages are available from the external GPS module

Each rising edge of the LiDAR's internal 1 Hz signal triggers a GPS Data Packet.
The time and date in the GPS Data Packets are unreal, starting from the UTC time 20 05 20 00 00 00 (year, month, day, hour, minute, second) and increasing with the internal 1 Hz signal.

Once the LiDAR receives the PPS (pulse-per-second) signal and NMEA messages

The internal 1 Hz signal will be locked to the PPS. Each rising edge still triggers a GPS Data Packet.
Meanwhile, the LiDAR will extract the actual date and time from NMEA messages ($GPRMC or $GPGGA), and stamp them into both Point Cloud Data Packets and GPS Data Packets.

- Point Cloud Data Packets: 6-byte Date & Time (year, month, day, hour, minute, second)
- GPS Data Packets: 6-byte Date (year, month, day) and 6-byte Time (second, minute, hour)

The GPS module sends first the PPS signal and then the NMEA message. At the rising edge of the PPS pulse, the corresponding NMEA message is not yet available. Therefore, the LiDAR extracts date and time from the previous NMEA message and automatically adds 1 full second.

When GPS signal is lost

The LiDAR will still trigger GPS Data Packets by the rising edge of the internal 1 Hz signal. However, the GPS time in the packets will be counted by the internal 1 Hz signal and will drift from the actual GPS time.

3.2.1 Ethernet Header

The source IP is 192.168.1.201 by default. The destination IP address is 255.255.255.255 and in broadcast form.

GPS Ethernet Header: 42 bytes		
Field	Bytes	Description
Ethernet II MAC	12	Destination: broadcast (0xFF: 0xFF: 0xFF: 0xFF: 0xFF: 0xFF) Source: (xx:xx:xx:xx:xx:xx)
Ethernet Data Packet Type	2	0x08, 0x00
Internet Protocol	20	Shown in the figure below
UDP Port Number	4	UDP source port (0x2710, represents 10000) Destination port (0x277E, represents 10110)
UDP Length	2	0x208, representing 520 bytes (8 bytes more than the size of the GPS UDP Data, shown in Figure 3.1)
UDP Checksum	2	-

```
⊟ Internet Protocol, Src: 192.168.1.201 (192.168.1.201), Dst: 255.255.255.255 (255.255.255.255)
    Version: 4
    Header length: 20 bytes
  ⊞ Differentiated Services Field: 0x00 (DSCP 0x00: Default: ECN: 0x00)
    Total Length: 540
    Identification: 0x1841 (6209)
  ⊞ Flags: 0x02 (Don't Fragment)
    Fragment offset: 0
    Time to live: 64
    Protocol: UDP (17)
  ⊞ Header checksum: 0x5e1f [correct]
    Source: 192.168.1.201 (192.168.1.201)
    Destination: 255.255.255.255 (255.255.255.255)
```

Figure 3.3 GPS Ethernet Header - Internet Protocol

3.2.2 UDP Data

<table>
<tr><th colspan="5">GPS UDP data: 512 bytes</th></tr>
<tr><th>Field</th><th>Bytes</th><th colspan="3">Description</th></tr>
<tr><td rowspan="4">GPS Time Data</td><td rowspan="4">18</td><td>Header</td><td>2 bytes</td><td>0xFFEE, 0xFF first</td></tr>
<tr><td>Date</td><td>6 bytes</td><td>Year, month, and day (2 bytes each, lower byte first) in ASCII</td></tr>
<tr><td>Time</td><td>6 bytes</td><td>Second, minute, and hour (2 bytes each, lower byte first) in ASCII</td></tr>
<tr><td>Reserved</td><td>4 bytes</td><td>-</td></tr>
<tr><td>GPRMC/GPGGA Data</td><td>84</td><td colspan="3">NMEA sentence that contains date and time
ASCII code, valid till 2 bytes after the asterisk (*)
The LiDAR can receive either GPRMC or GPGGA, see Chapter 4 (Web Control - Settings)</td></tr>
<tr><td>Reserved</td><td>404</td><td colspan="3">404 bytes of 0xDF</td></tr>
<tr><td>GPS Positioning Status</td><td>1</td><td colspan="3">ASCII code, obtained from $GPRMC or $GPGGA

When $GPRMC is selected:
A (hex = 41) for Valid Position
V (hex = 56) for Invalid Position
NUL (hex = 0) for GPS being unlocked

When $GPGGA is selected:
0 = invalid
1 = GPS fix (SPS)
2 = DGPS fix
3 = PPS fix
6 = estimated (dead reckoning)</td></tr>
<tr><td>PPS Lock Flag</td><td>1</td><td colspan="3">1 - locked 0 - unlocked</td></tr>
<tr><td>Reserved</td><td>4</td><td colspan="3">-</td></tr>
</table>

GPRMC Data Format

$GPRMC, <01>, <02>, <03>, <04>, <05>, <06>, <07>, <08>, <09>, <10>, <11>, <12>*hh

Field #	Field	Description
<01>	UTC Time	Hour, minute, and second Typically in hhmmss (hour, minute, second) format
<02>	Location Status	A (hex = 41) for Valid Position V (hex = 56) for Invalid Position NUL (hex = 0) for GPS being unlocked
...		
<09>	UTC Date	Date information Typically in ddmmyy (day, month, year) format
...		

The LiDAR's GPS data interface is compatible with a variety of GPRMC formats, as long as:

<01> is the hour, minute, and second information
<09> is the date information.

For example, the following two formats are both acceptable:

$GPRMC,072242,A,3027.3680,N,11423.6975,E,000.0,316.7,160617,004.1,W*67
$GPRMC,065829.00,A,3121.86377,N,12114.68322,E,0.027,,160617,,,A*74

GPGGA Data Format

$GPGGA, <01>, <02>, <03>, <04>, <05>, <06>, <07>, <08>, <09>, <10>, <11>, <12>*hh

Field #	Field	Description
<01>	UTC Time	Hour, minute, and second Typically in hhmmss (hour, minute, second) format
...		
<06>	GPS Fix Quality	0 = invalid 1 = GPS fix (SPS) 2 = DGPS fix 3 = PPS fix 6 = estimated (dead reckoning)
...		

The LiDAR's GPS data interface is compatible with a variety of GPGGA formats, as long as:

<01> is the hour, minute, and second information

For example, the following two formats are both acceptable:

$GPGGA,123519,4807.038,N,01131.000,E,1,08,0.9,545.4,M,46.9,M,,*47
$GPGGA,134658.00,5106.9792,N,11402.3003,W,2,09,1.0,1048.47,M,-6.27,M,08,AAAA*60

3.2.3 GPS Data Analysis

```
> Data (512 bytes)
0000  04 d4 c4 eb 9b 37 ec 9f  0d 00 48 cb 08 00 45 00   ·····7·· ··H···E·
0010  02 1c c4 23 40 00 80 11  b0 66 c0 a8 01 c9 c0 a8   ···#@··· ·f······
0020  01 2d 27 10 27 7e 02 08  00 00 ff ee 30 32 34 30   ·-'·'~·· ····0240
0030  37 30 38 35 37 30 34 30  00 00 00 00 24 47 50 52   70857040 ····$GPR
0040  4d 43 00 2c 30 34 30 37  35 37 2e 37 36 2c 56 2c   MC·,0407 57.76,V,
0050  2c 2c 2c 2c 2c 2c 30 37  30 34 32 30 2c 2c 2c 4e   ,,,,,,07 0420,,,N
0060  2c 56 2a 30 36 36 36 36  36 36 36 36 36 36 36 36   ,V*06666 66666666
```

Figure 3.3 GPS Data Packet - UDP Data (Example)

Date

Field	Data (ASCII Code)	Characters	Meaning
Year	0x30 0x32	'0', '2'	20
Month	0x34 0x30	'4', '0'	04
Day	0x37 0x30	'7', '0'	07

Time

Field	Data (ASCII Code)	Characters	Meaning
Second	0x38 0x35	'8', '5'	58
Minute	0x37 0x30	'7', '0'	07
Hour	0x34 0x30	'4', '0'	04

μs Time

4 bytes, in units of μs, using the same clock source as the GPS Timestamp in Point Cloud Data Packets

Reset to 0 at the rising edge of each PPS signal

4 Web Control

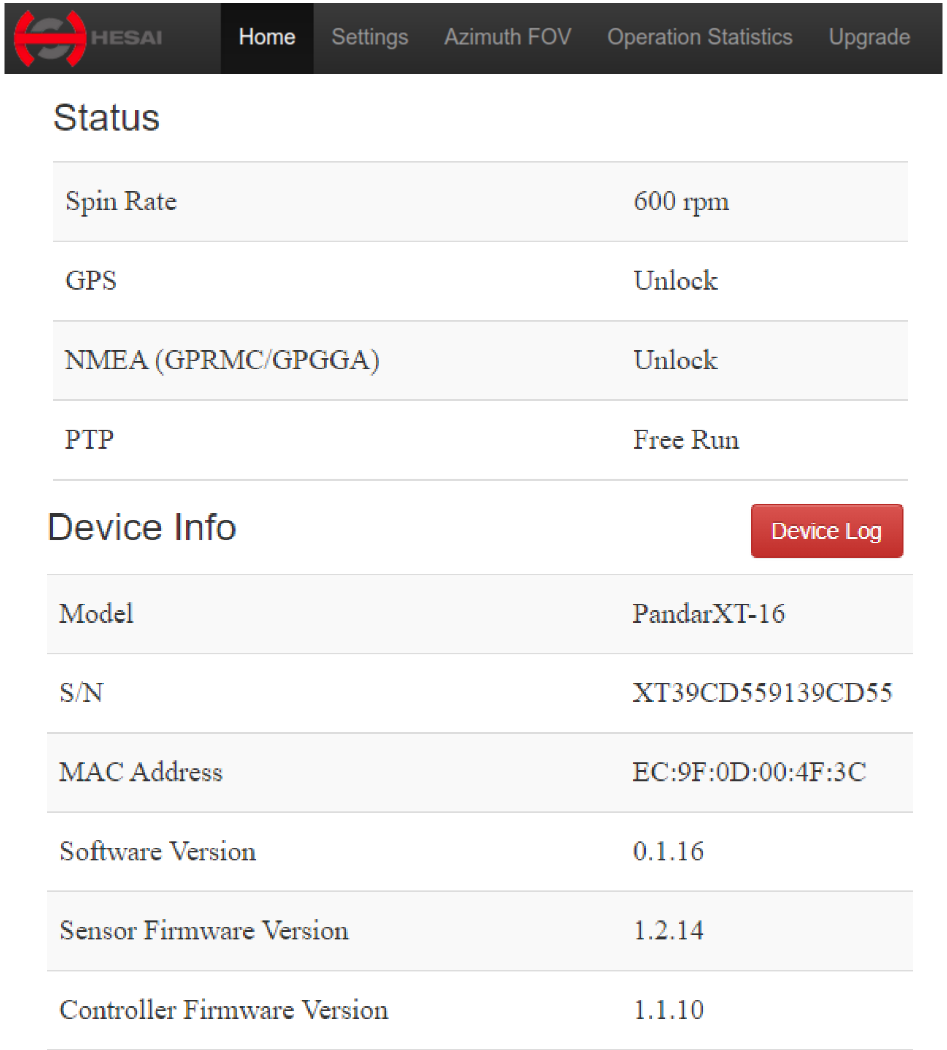
Web control is used for setting parameters, checking device info, and upgrading.

To access web control

1) Connect the LiDAR to your PC using an Ethernet cable
2) Set the IP address according to Section 2.4 (Get Ready to Use)
3) Enter this URL into your web browser: 192.168.1.201/index.html

NOTE Google Chrome and Mozilla Firefox are recommended.

4.1 Home



NOTE The screenshot may not display the most current version numbers.

Spin Rate of the motor (revs per minute) = frame rate (Hz) * 60

GPS (PPS) Status

Lock	LiDAR's internal clock is in sync with GPS
Unlock	Not in sync

NMEA (GPRMC/GPGGA) Status

Lock	After receiving a valid NMEA message
Unlock	No valid NMEA message for over 2 seconds

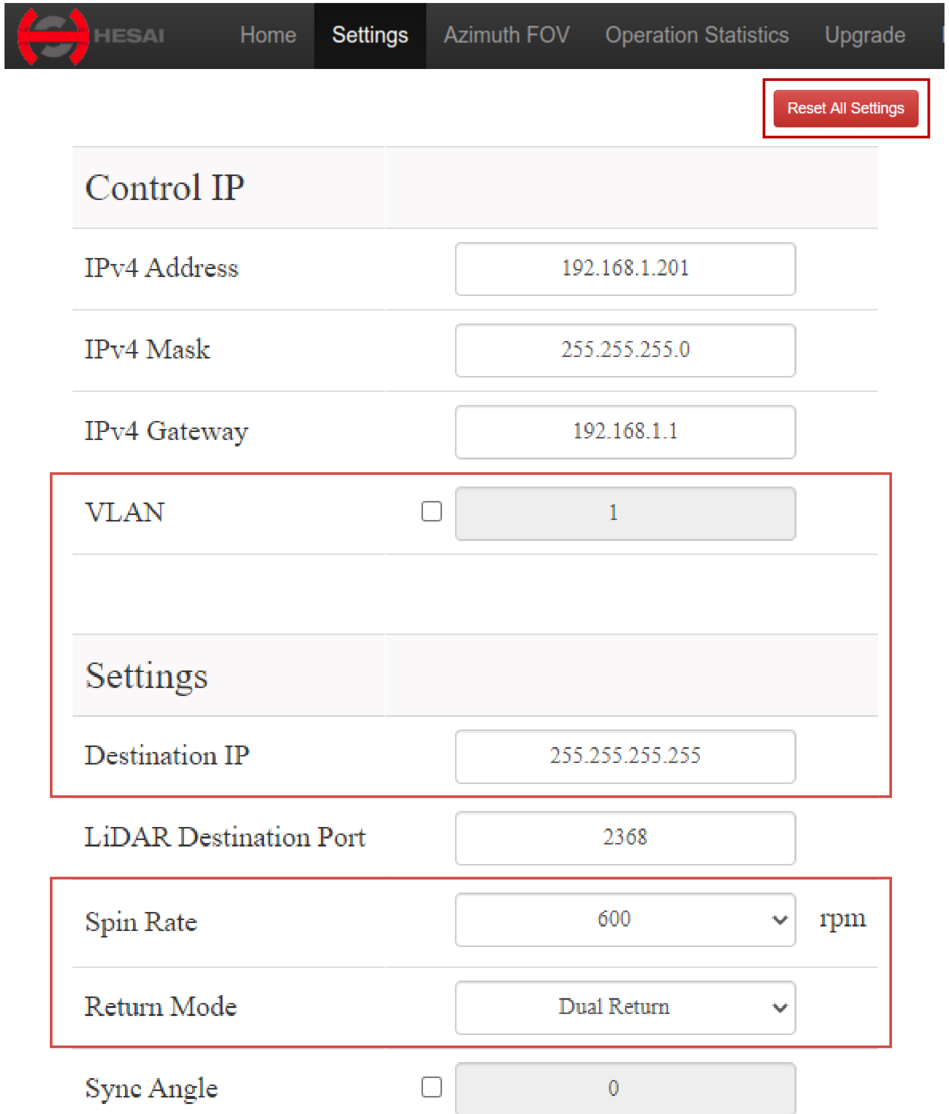
PTP Status

Free Run	No PTP master is selected
Tracking	Slave is trying to sync with the selected PTP Master, but the absolute offset is over 1 μs
Locked	Absolute offset between Slave and Master is < 1 μs
Frozen (Holdover)	LiDAR has lost connection to the PTP master and is attempting to recover it. Meanwhile, LiDAR starts drifting from the previous clock; when drifting out of specifications, it goes back to the Free Run mode.

Device Log

Click to download a .JSON file containing the LiDAR's status, device info, all configurable parameters, and upgrade log.

4.2 Settings



(continued on the next page)

1. Reset All Settings

By clicking the "Reset All Settings" button on the top-right corner, all configurable parameters on web control will be reset to factory defaults.

The default values are shown in the screenshots in Section 4.2 (Settings) and Section 4.3.1 (Azimuth FOV - for All Channels).

2. Control IP

VLAN Tagging can be used when the receiving host also supports VLAN function.

- Check the VLAN checkbox and input a VLAN ID (1~4094)
- Set the VLAN ID of the receiving host to be the same

3. Destination IP

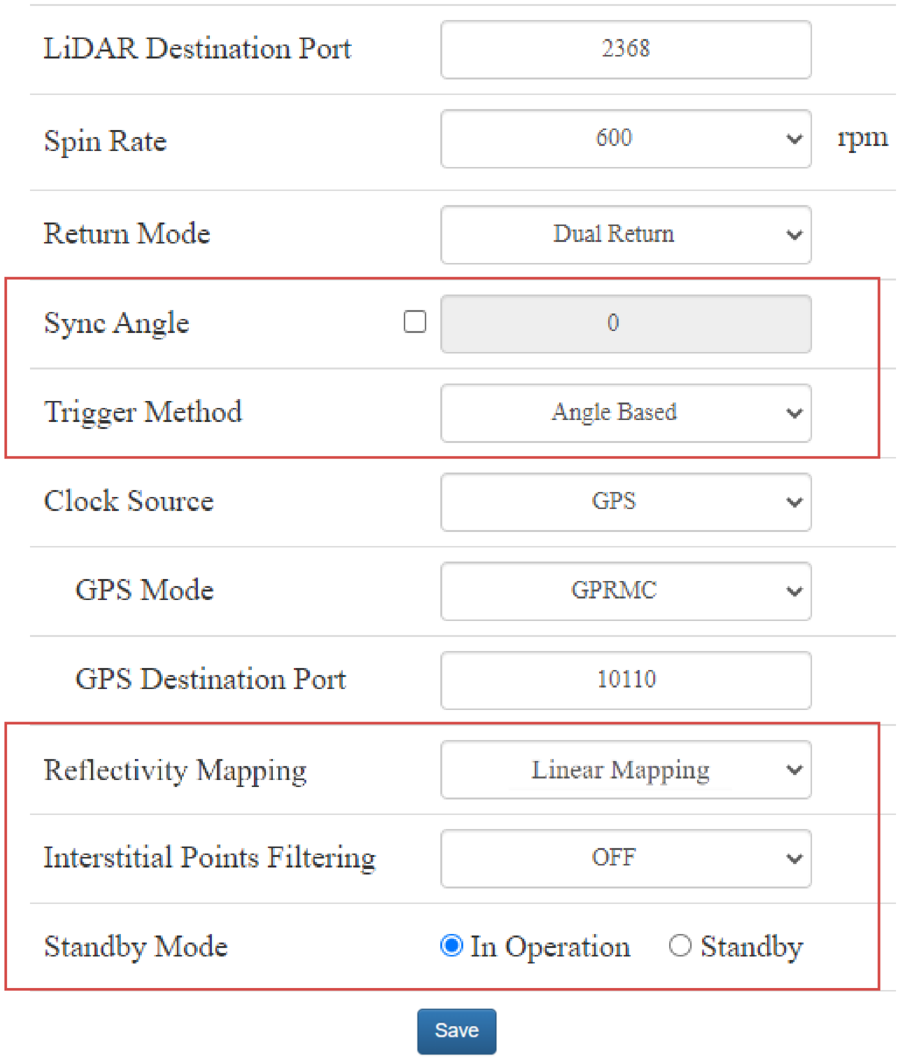
Range: except for 0.0.0.0, 127.0.0.1, and the LiDAR's IP

Mode	Destination IP
Broadcast (default)	255.255.255.255
Multicast	239.0.0.0~239.255.255.255
Unicast	Same as the PC's IP address

4. LiDAR Functions

Spin Rate	600 rpm / 1200 rpm
Return Mode	Last / Strongest / Dual Return

(continued)

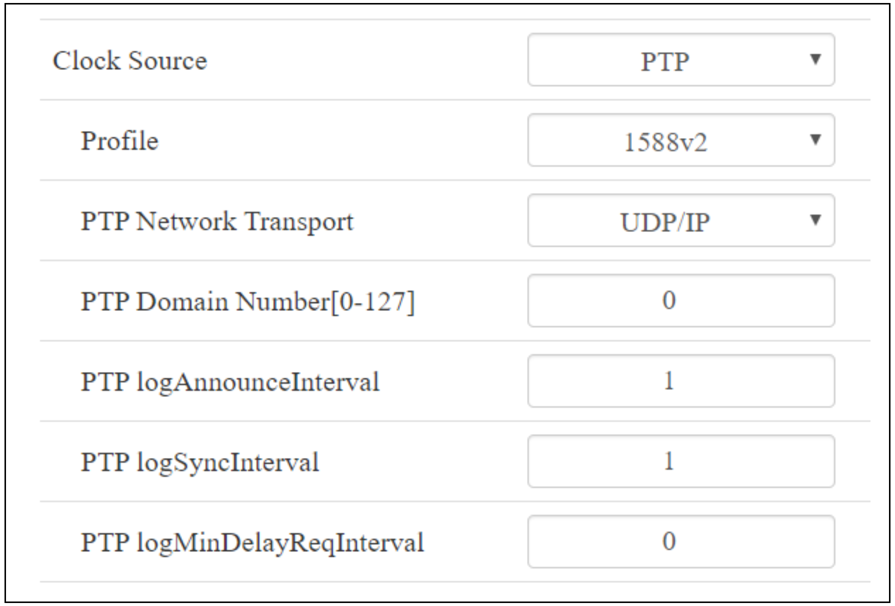


Sync Angle	0~360 degrees
	By default, the LiDAR's 0° position (see Section 1.2) is not in sync with GPS PPS or the whole second of the PTP clock. If syncing is needed, check the checkbox and input a sync angle.
Trigger Method	Angle-Based / Time-Based
	Angle-based: lasers fire every 0.09° at 5 Hz, 0.18° at 10 Hz, or 0.36° at 20 Hz. **Time-based**: lasers fire every 50 us.
Reflectivity Mapping	Linear / Nonlinear Mapping
	Linear: the 1-byte reflectivity in Point Cloud Data Packets linearly represents target reflectivity (0 to 255%). **Nonlinear**: increases the contrast in low-reflectivity region, see the appendix.
Interstitial Points Filtering	Interstitial point: when a beam partially hits on a front target's edge and further hits on a rear target, the return signal can result in a false point located between both targets. Such points can be mitigated.
Standby Mode	In Operation / Standby
	In Standby mode, the motor stops running and lasers stop firing.

(continued on the next page)

5. Clock Source and PTP Parameters

Clock Source	GPS / PTP
	In PTP mode, the LiDAR does not output GPS Data Packets



- When GPS is selected as the clock source:

GPS Mode	GPRMC / GPGGA
	Format of NMEA data received from the external GPS module, see Section 3.2.2
GPS Destination Port	10110 (default)
	Port used for sending GPS Data packets

- When PTP is selected as the clock source:

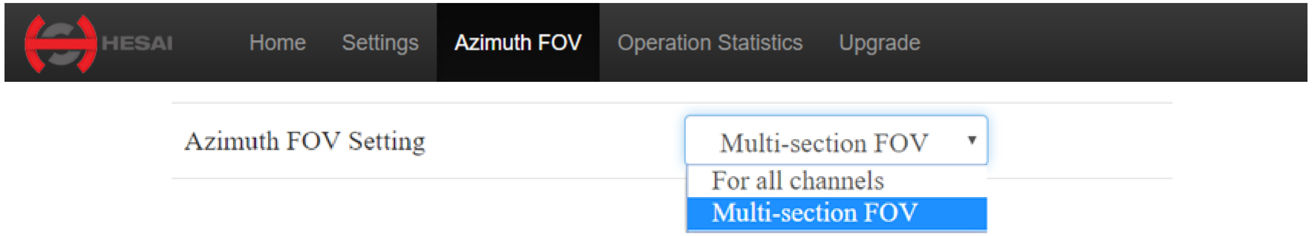
Profile	1588v2 (default) / 802.1AS
	IEEE timing and synchronization standard
PTP Network Transport	UDP/IP (default) or L2
	1588v2: users can select UDP/IP or L2 **802.1AS**: only supports L2 network
PTP Domain Number	Integer from 0 to 127
	Domain attribute of the local clock

- When using the 1588v2 profile:

PTP logAnnounceInterval	-2 to 3 log seconds
	Time interval between Announce messages (default: 1)
PTP logSyncInterval	-7 to 3 log seconds
	Time interval between Sync messages (default: 1)
PTP logMinDelayReq-Interval	-7 to 3 log seconds
	Minimum permitted mean time between Delay_Req messages (default: 0)

4.3 Azimuth FOV

To set the Azimuth FOV, users can select one of the two modes: for all channels, or multi-section FOV.

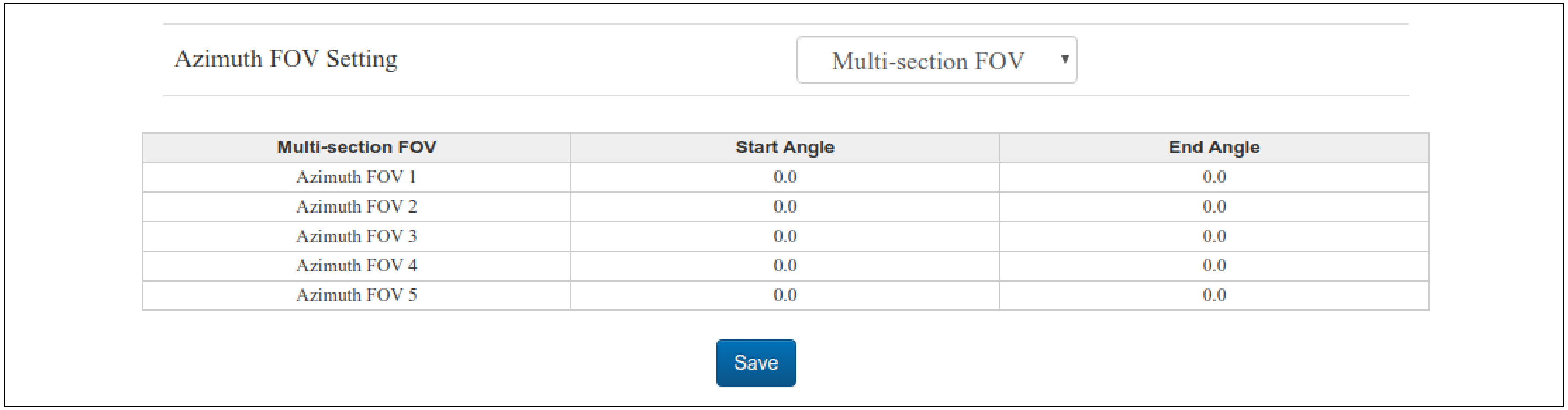


4.3.1 For all channels

A continuous angle range, specified by a Start Angle and an End Angle, will be applied to all channels. The LiDAR outputs valid data only within the specified range.

4.3.2 Multi-section FOV

Users can configure up to five continuous angle ranges for all channels.
Each channel outputs valid data only within its specified range.



Multi-section FOV	Start Angle	End Angle
Azimuth FOV 1	0.0	0.0
Azimuth FOV 2	0.0	0.0
Azimuth FOV 3	0.0	0.0
Azimuth FOV 4	0.0	0.0
Azimuth FOV 5	0.0	0.0

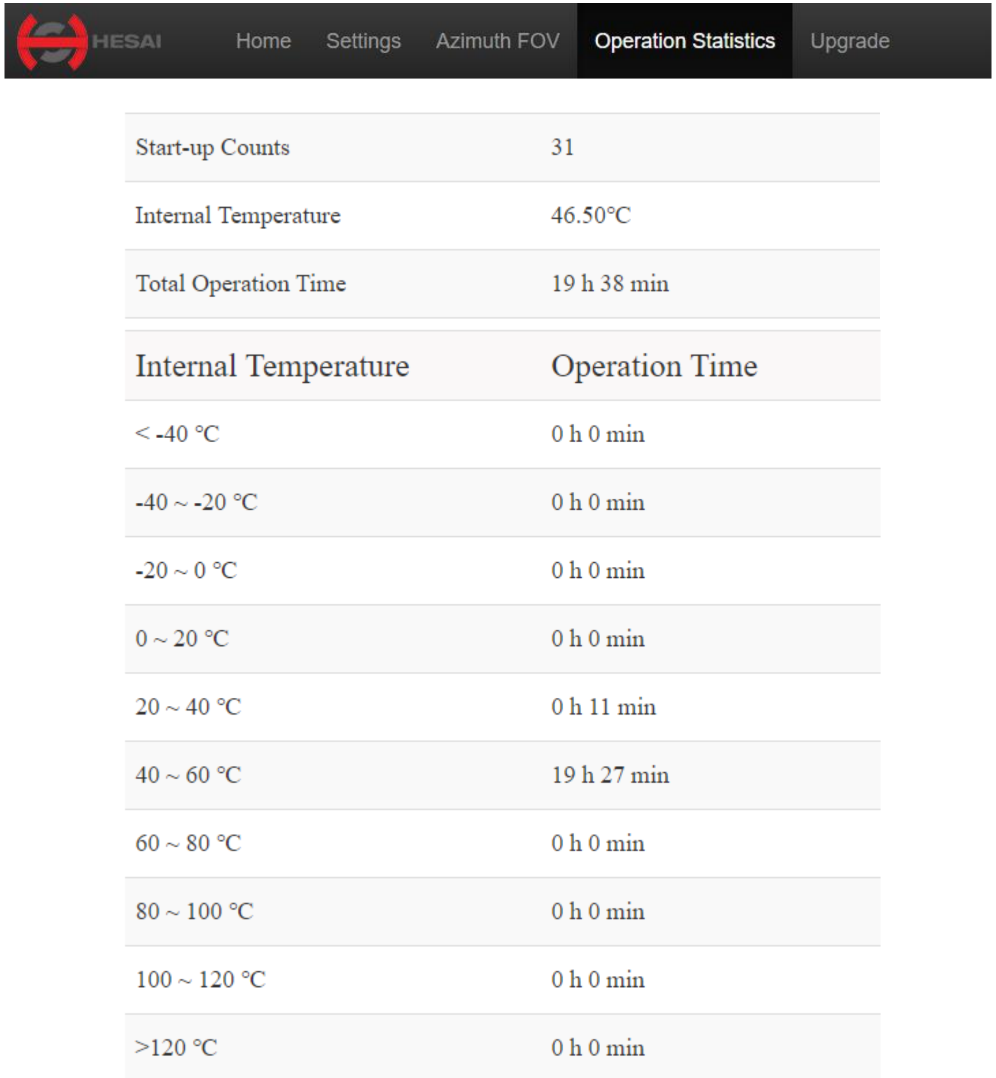
4.3.3 Note

- Click "Save" to apply your settings.
- The angles in degrees are accurate to the first decimal place.
- If the Start Angle is larger than the End Angle, then the actual azimuth FOV is the union of [Start Angle, 360°] and [0°, End Angle].

For instance, when the angle range is set to be [270°, 90°], the actual azimuth FOV is [270°, 360°]∪[0°, 90°].

4.4 Operation Statistics

The LiDAR's operation time in aggregate and in different temperature ranges are listed.

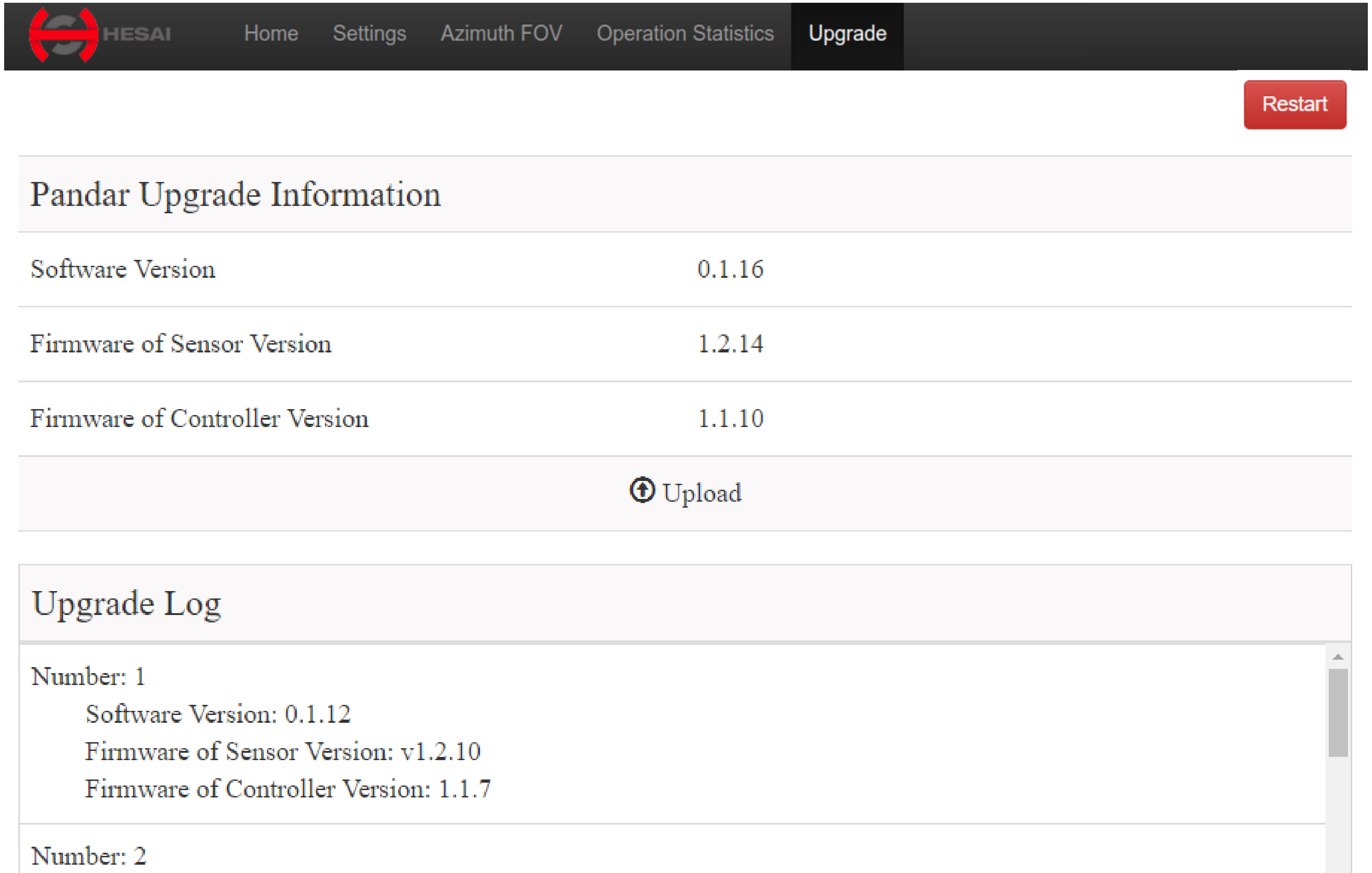


Start-up Counts	31
Internal Temperature	46.50°C
Total Operation Time	19 h 38 min

Internal Temperature	Operation Time
< -40 °C	0 h 0 min
-40 ~ -20 °C	0 h 0 min
-20 ~ 0 °C	0 h 0 min
0 ~ 20 °C	0 h 0 min
20 ~ 40 °C	0 h 11 min
40 ~ 60 °C	19 h 27 min
60 ~ 80 °C	0 h 0 min
80 ~ 100 °C	0 h 0 min
100 ~ 120 °C	0 h 0 min
>120 °C	0 h 0 min

4.5 Upgrade

Click the "Upload" button, select an upgrade file (provided by Hesai), and confirm your choice in the pop-up window.
When the upgrade is complete, the LiDAR will automatically reboot, and the past versions will be logged in the Upgrade Log.



A software reboot is triggered by clicking the "Restart" button on the top right corner.

Afterwards, the start-up counts in the Operation Statistics page increments by 1.

NOTE The screenshot may not display the most current version numbers.

5 PandarView

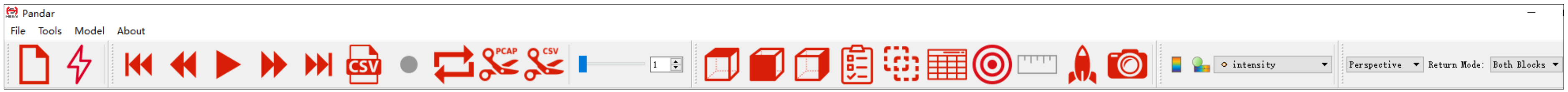
PandarView is a software that records and displays point cloud data from Hesai LiDARs, available in 64-bit Windows 10 and Ubuntu-16.04/18.04.

5.1 Installation

Copy the installation files from the USB disk in the LiDAR's protective case, or download these files from Hesai's official website:
www.hesaitech.com/en/download

System	Installation Files	Installation Steps
Windows	PandarViewX64_Release_V1.7.37.msi	Before upgrading PandarView to a newer version, please uninstall the current version
		Double click and install PandarView_Windows using the default settings
Ubuntu-16.04	PandarViewX64_Release_V1.7. 37.tar.gz	Unzip the file and run PandarView_Installer.bin
Ubuntu-18.04	PandarViewX64_18.04_Release_V1.7. 37.tar.gz	

This manual describes PandarView 1.7.37. The menu bar and buttons are shown below.

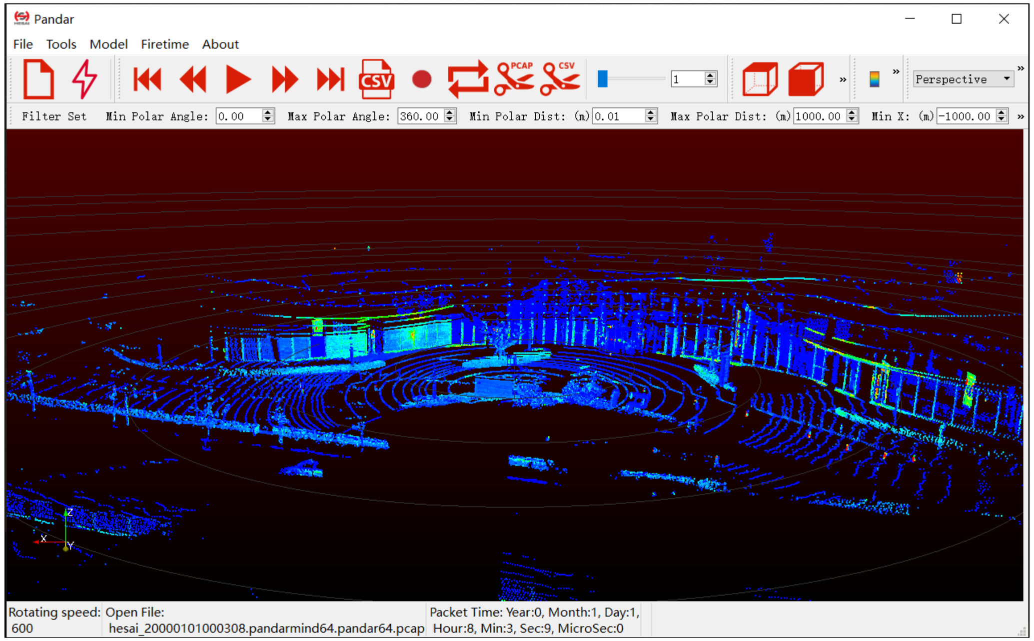


NOTE Users may check the software version from "About" in the menu bar.

5.2 Check Live Data

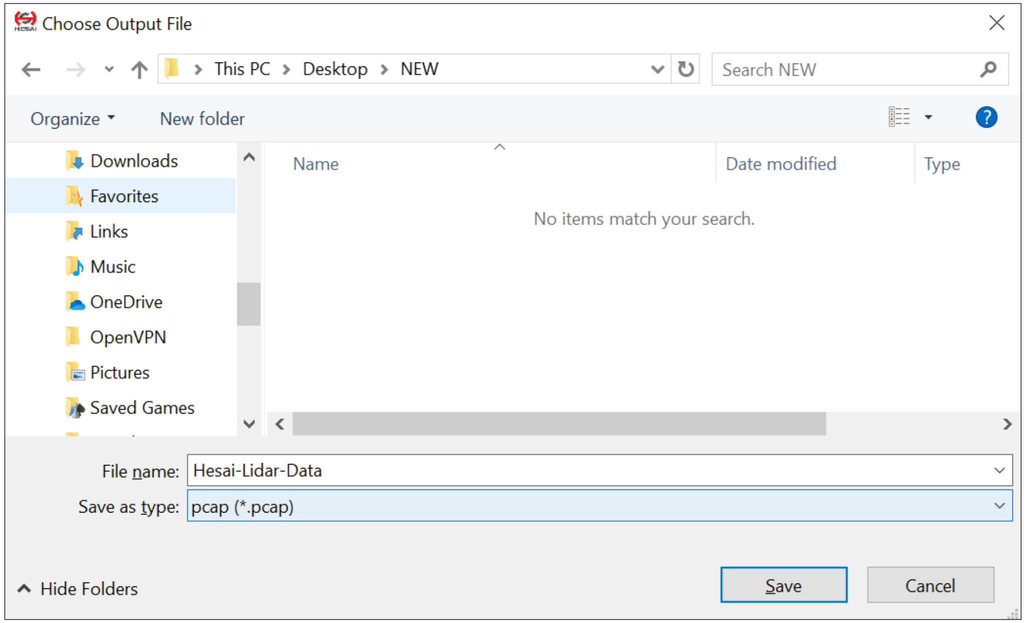
Set the PC's IP address according to Section 2.4 (Get Ready to Use)

Click on and select your LiDAR model to begin receiving data over Ethernet.



5.3 Record Point Cloud Data

1) Click on to pop up the "Choose Output File" window.
2) Specify the file directory and click on "Save" to begin recording a .PCAP file.
3) Click on again to stop recording.



5.4 Play Point Cloud Data

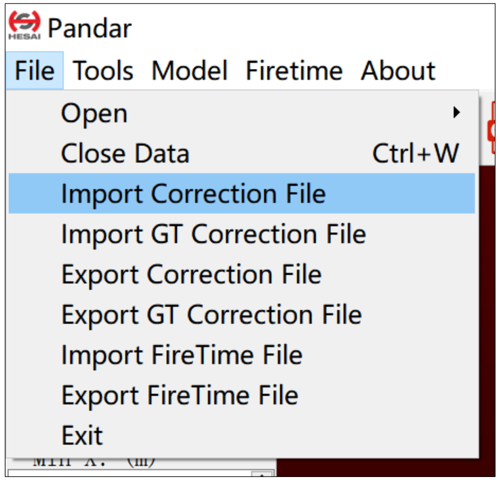
1) Open a .PCAP File

Click on to pop up the "Choose Open File" window. Select a .PCAP file to open.

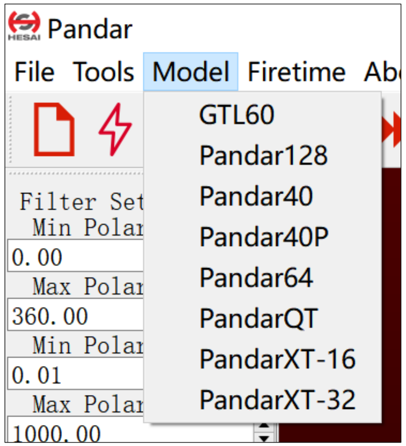
2) Import a Correction File

Each LiDAR unit has a corresponding calibration file (.CSV), see Section 1.3 (Channel Distribution).

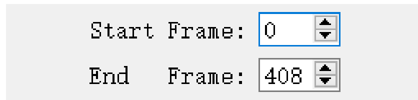
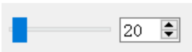
We recommend importing the calibration file of this LiDAR unit into PandarView (File -- Import Correction File), in order to display the point cloud most accurately.



If the calibration file of this LiDAR unit is temporarily not at hand, select the LiDAR model in the "Model" menu. Thus a general calibration file for this model will be loaded to improve point cloud display.



3) Play the .PCAP File

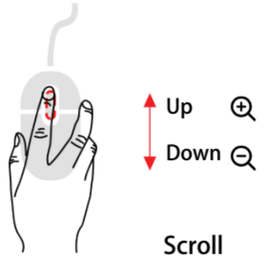
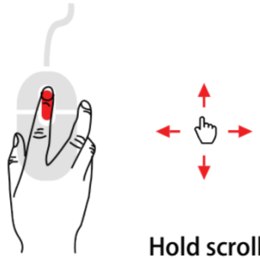
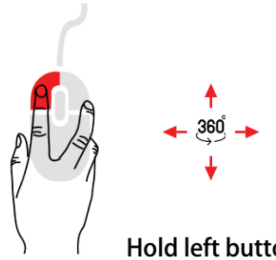
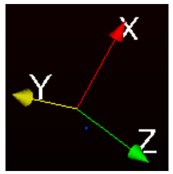
Button	Description	
	Jump to the beginning of the file	
	While paused, jump to the previous frame While playing, rewind. May click again to adjust the rewind speed (2x, 3x, 1/2x, 1/4x, and 1x)	
 / 	After loading a point cloud file, click to play the file While playing, click to pause	
	While paused, jump to the next frame. While playing, forward. May click again to adjust the forward speed (2x, 3x, 1/2x, 1/4x, and 1x)	
	Jump to the end of the file	
	Save a single frame to .CSV	
	While playing, this Record button will be gray and unclickable	
	While playing, click to loop playback. Otherwise the player will stop at the end of the file	
	Save multiple frames to .PCAP	Specify the start and end frames
	Save multiple frames to .CSV	
	Drag this progress bar or enter a frame number to jump to a specific frame	

5.5 Features

■ Standard Viewpoints

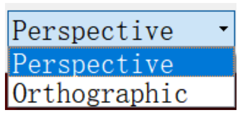
		
Right	Front	Top

■ Mouse Shortcuts

 Scroll	 Hold scroll	 Hold left button	
Scroll the mouse wheel up/down to zoom in/out	Press the mouse wheel and drag to pan the view	Hold the left button and drag to adjust the point of view	The bottom-left coordinate axes show the current point of view

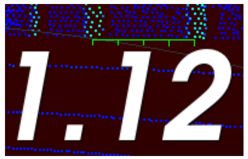
■ 3D Projection and Distance Measurement

PandarView supports perspective projection (default) and orthographic projection.



The distance ruler is available only under orthographic projection:

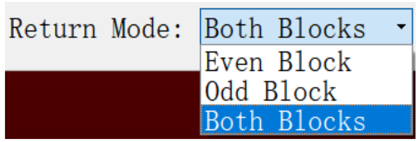
- Click on to enter measurement mode. Hold the Ctrl key and drag the mouse to make a measurement in units of meters
- Click on again to quit



Return Mode

- Both blocks (default): to show the point cloud data from all blocks
- Even/Odd Block: to show the point cloud data from even/odd-number blocks

NOTE See the definition of blocks in Section 3.1.2 (Point Cloud UDP Data)



UDP Port

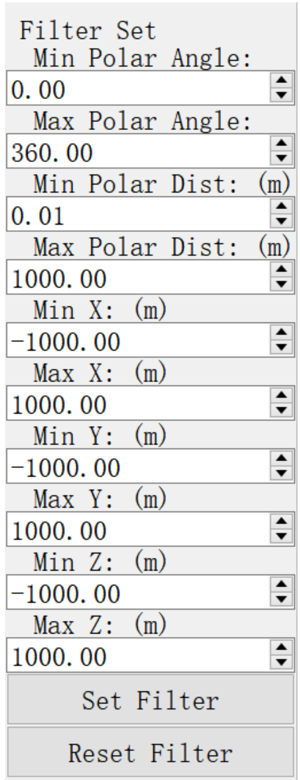
Enter the UDP port number and click "Set".

View Filter

To set the polar/rectangular coordinate range for viewing live point cloud data or a .PCAP file.

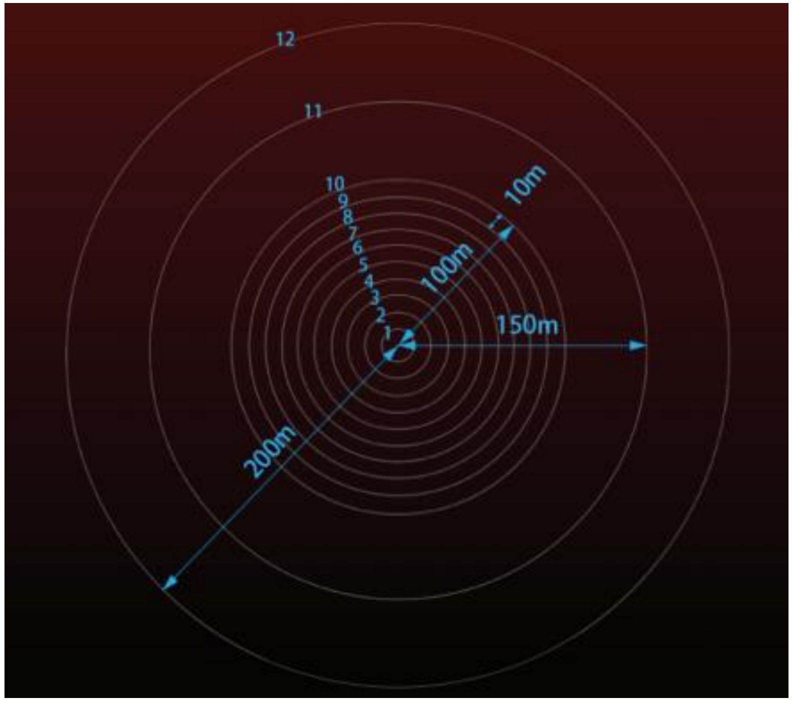
- Click "Set Filter" to apply the settings.
- Click "Reset Filter" to return to default settings (shown in the screenshot).

NOTE The filter does not apply to recording and saving .PCAP files.



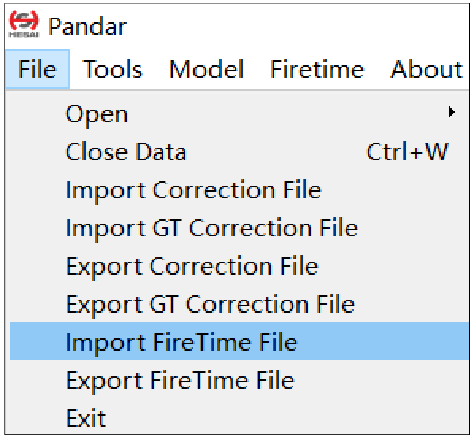
Distance Reference Circles

Click on to show/hide the 12 distance reference circles. The actual distances are marked below.
To change the color and line width of these circles, click on "Tools" in the menu bar and open "Grid Properties".



Fire Time Correction

After opening a .PCAP file, import the fire time correction file of this LiDAR model into PandarView (File -- Import FireTime File).

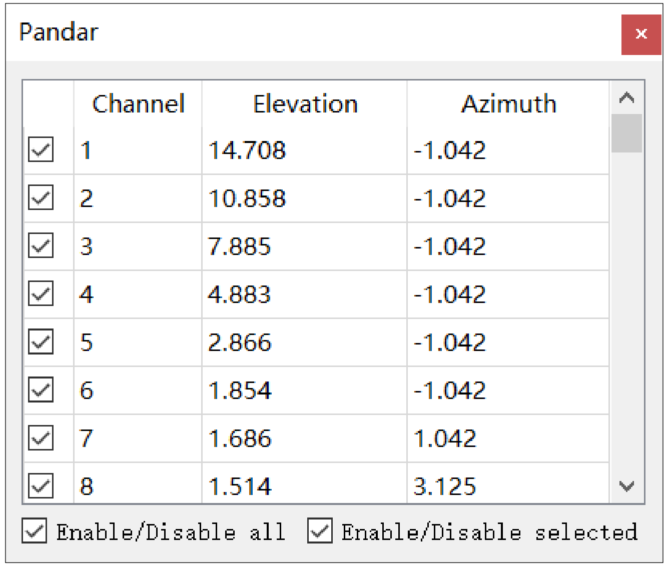


Afterwards, click on to finetune point cloud display using the fire time correction file.

Click on again to cancel the finetuning effects.

Channel Selection

Click on to open the Channel Selection box.

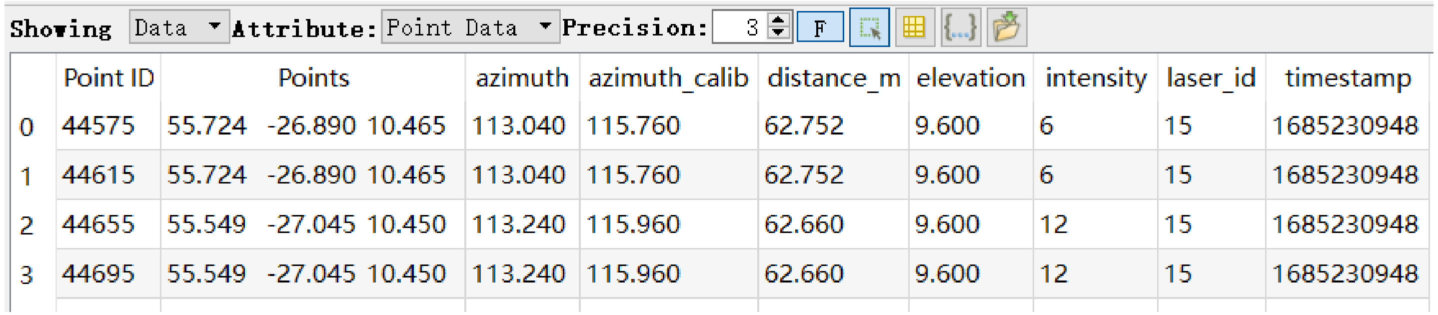


- Check/Uncheck the boxes on the left to show/hide each channel. By default, the point cloud data from all channels are shown.
- Check/Uncheck the "Enable/Disable all" option at the bottom of the table to show/hide all channels.
- When multiple channels are selected by holding the Shift or Ctrl key, check/uncheck the "Enable/Disable selected" option to show/hide multiple channels.

Click on again to close the Channel Selection box.

Point Selection and Data Table

Click on and drag the mouse over the point cloud to highlight an area of points.
Click on to view the data of the highlighted points, as shown below.



	Point ID	Points	azimuth	azimuth_calib	distance_m	elevation	intensity	laser_id	timestamp
0	44575	55.724 -26.890 10.465	113.040	115.760	62.752	9.600	6	15	1685230948
1	44615	55.724 -26.890 10.465	113.040	115.760	62.752	9.600	6	15	1685230948
2	44655	55.549 -27.045 10.450	113.240	115.960	62.660	9.600	12	15	1685230948
3	44695	55.549 -27.045 10.450	113.240	115.960	62.660	9.600	12	15	1685230948

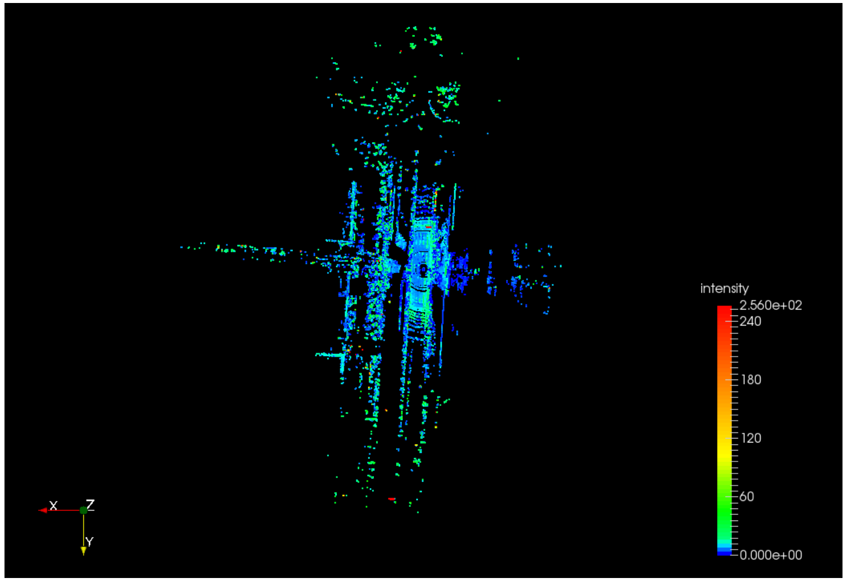
Some of the data fields are defined below:

Field	Description
points	The XYZ coordinates of each point
azimuth	Rotor's current reference angle
azimuth_calib	Azimuth + horizontal angle offset

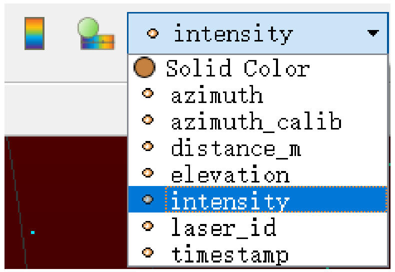
To cancel the selection, click on again and click on any place outside the selected point cloud area.

Color Schemes

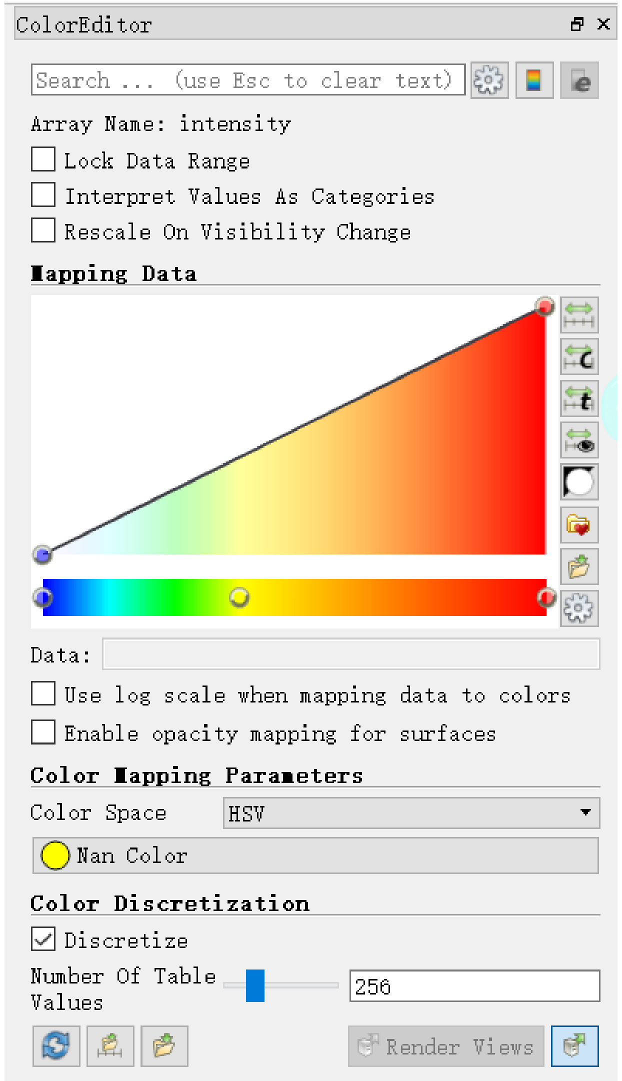
Click on to show the color legend at the lower right corner.



The default color scheme is intensity based. Users can choose from other colors schemes based on azimuth, azimuth_calib, distance, elevation, laser_id, or timestamp.



Click on to open or close the Color Editor.



6 Communication Protocol

To receive Hesai LiDAR's PTC (Pandar TCP Commands) and HTTP API Protocols, please contact Hesai technical support.

7 Sensor Maintenance

■ Storage

The product has passed the high- and low-temperature storage tests in ISO 16750, in which the test temperature range is -40°C to 85°C.
We recommend storing the product in a dry, well ventilated place, under room temperature (23±5°C) and a relative humidity of 30% to 70%.
Please check Section 1.4 (Specifications) for the IP rating, and avoid any ingress beyond that rating.

■ Transport

Package the product in shock-proof materials to avoid damage during transport.

■ Cleaning

Stains on the product's enclosure, such as dirt, fingerprints, and oil, can negatively affect point cloud data quality.
Please perform the following steps to remove the stains.

NOTE

- To avoid damaging the optical coating, DO NOT apply pressure when wiping the enclosure
- Only clean the stained area of the enclosure
- Check before using a lint-free wipe. If the wipe is stained, use another

1) Thoroughly wash your hands or wear a pair of powder-free PVC gloves

2) To remove dust, blow dry air onto the enclosure, or use a piece of lint-free wipe to lightly brush across the dusty area

To remove persistent stains, move on to the next step

(Continued on the next page)

(Continued)

3) Spray the enclosure with warm, neutral solvent using a spray bottle

Solvent type	99% isopropyl alcohol (IPA) or 99% ethanol (absolute alcohol) or distilled water **NOTE** When using IPA or alcohol, please ensure adequate ventilation and keep away from fire.
Solvent temperature	20 to 25°C

4) When the stains have loosened, dip a piece of lint-free wipe into the solvent made in Step 3, and gently wipe the enclosure back and forth along its curved surface

5) Should another cleaning agent be applied to remove certain stains, repeat Steps 3 and 4

6) Spray the enclosure with clean water, and gently wipe off the remaining liquid with another piece of lint-free wipe

8 Troubleshooting

In case the following procedures cannot solve the problem, please contact Hesai technical support.

Symptoms	Points to Check
Indicator light is off on the connection box	Verify that • power adapter is properly connected and in good condition • connection box is intact • input voltage and current satisfy the requirements in Section 2.3 (Connection Box) Power on again to check if the symptom persists.
Motor is not running	Verify that • power adapter is properly connected and in good condition • if a connection box is used, the connection box is intact • input voltage and current satisfy the requirements in Section 1.4 (Specifications) and 2.3 (Connection Box) • web control can be accessed (see "cannot open web control" on the next page) Power on again to check if the symptom persists.
Motor is running but no output data is received, neither on Wireshark nor on PandarView	Verify that • Ethernet cable is properly connected (by unplugging and plugging again) • LiDAR's IP is in the same subnet with the PC's • horizontal FOV is properly set on the Azimuth FOV page of web control • firmware version of the sensor is correctly shown on the Upgrade page of web control • LiDAR is emitting laser light. This can be checked by using an infrared camera, an infrared sensor card, or a phone camera without infrared filter Power on again to check if the symptom persists.

(Continued on the next page)

(Continued)

Symptoms	Points to Check
Can receive data on Wireshark but not on PandarView	Verify that ・ Destination IP and the Destination LiDAR Port are correctly set on the Settings page of web control ・ PC's firewall is disabled, or that PandarView is added to the firewall exceptions ・ the latest PandarView version (see the Download page of Hesai's official website) is installed on the PC Power on again to check if the symptom persists.
Cannot open web control	Verify that ・ Ethernet cable is properly connected (by unplugging and plugging again) ・ LiDAR's IP is in the same subnet with the PC's. Users may use WireShark to check the LiDAR's IP that broadcasts data packets Afterwards, ・ restart PC, or connect the LiDAR to another PC ・ power on again to check if the symptom persists
Abnormal packet size (missing packets)	Verify that ・ horizontal FOV is properly set on the Azimuth FOV page of web control ・ motor's spin rate is steady on the Home page of web control ・ LiDAR's internal temperature is between -20°C and 95°C on the Operation Statistics page of web control ・ Ethernet is not overloaded ・ no switch is connected into the network. The data transmitted from other devices may cause network congestion and packet loss Afterwards, ・ connect the PC only to the LiDAR and check for packet loss ・ power on again to check if the symptom persists

(Continued on the next page)

(Continued)

Symptoms	Points to Check
Abnormal point cloud (obviously misaligned points, flashing points, or incomplete FOV)	Verify that • LiDAR's enclosure is clean. If not, refer to Chapter 7 (Sensor Maintenance) for the cleaning method • LiDAR's calibration file is imported, see Section 5.2 (PandarView - Use) • horizontal FOV is properly set on the Azimuth FOV page of web control • motor's spin rate is steady on the Home page of web control • LiDAR's internal temperature is between -20°C and 95°C on the Operation Statistics page of web control Afterwards, check for packet loss • If no packet is missing while the point cloud flashes, please update PandarView to the latest version (see the Download page of Hesai's official website) and restart the PC If the point cloud is still abnormal • Try connecting the LiDAR to another PC • Power on again to check if the symptom persists
GPS cannot be locked	Verify that • GPS receiver is properly connected • PPS signal is connected to the LiDAR • Destination GPS Port is correct on the Settings page of web control • input GPS signals satisfy the electrical requirements in Section 2.2 (Interface) and Section 2.3.1 (Connection Box) Power on again to check if the symptom persists

Appendix I Channel Distribution

The Vertical Angles (Elevation) in the table below are design values.
The accurate values are in this LiDAR's unit's calibration file, see Section 1.3 (Channel Distribution) and Section 3.1.3 (Point Cloud Data Analysis).

PandarXT-16 Channel Distribution

Channel # in UDP Data	Horizontal Angle Offset (Azimuth)	Vertical Angle (Elevation)	Instrument Range (in meters)	Range (in meters) with Reflectivity
01 (Top)	0°	15°	120	50@10%
02	0°	13°	120	50@10%
03	0°	11°	120	50@10%
04	0°	9°	120	50@10%
05	0°	7°	120	80@10%
06	0°	5°	120	80@10%
07	0°	3°	120	80@10%
08	0°	1°	120	80@10%
09	0°	-1°	120	80@10%
10	0°	-3°	120	80@10%
11	0°	-5°	120	80@10%
12	0°	-7°	120	80@10%
13	0°	-9°	120	50@10%
14	0°	-11°	120	50@10%
15	0°	-13°	120	50@10%
16 (Bottom)	0°	-15°	120	50@10%

Appendix II Absolute Time and Laser Firing Time

■ Absolute Time of Point Cloud Data Packets

The Body of each Point Cloud Data Packet contains 8 data blocks, as detailed in Section 3.1.2 (Point Cloud UDP Data).

Single Return Mode

The measurements from one round of firing are stored in one block.
The absolute time of a Point Cloud Data Packet is the time when the LiDAR sends a command to trigger a round of firing that will be stored in Block 8.

Dual Return Mode

The measurements from one round of firing are stored in two adjacent blocks, see Section 3.1.2 (Point Cloud UDP Data).
The absolute time of a Point Cloud Data Packet is the time when the LiDAR sends a command to trigger a round of firing that will be stored in Blocks 7 & 8.

Calculation

The absolute time of a Point Cloud Data Packet is calculated as the sum of date, time (accurate to the second) and μs time.

- Date and Time can be retrieved either from the current Point Cloud Data Packet (6 bytes of Date & Time), or from the previous GPS Data Packet (6 bytes of Date and 6 bytes of Time).
- μs time can be retrieved from the current Point Cloud Data Packet (4 bytes of Timestamp)

NOTE When using a PTP clock source, the LiDAR does not output GPS Data Packets.

■ Start Time of Each Block

Assuming that the absolute time of a Point Cloud Data Packet is t0, the start time of each block (the time when the first firing starts) can be calculated.

Single Return Mode

Block	Start Time (μs)
Block 8	t0 + 3.28
Block N	t0 + 3.28 - 50 * (8 - N)
Block 3	t0 + 3.28 - 50 * 5
Block 2	t0 + 3.28 - 50 * 6
Block 1	t0 + 3.28 - 50 * 7

Dual Return Mode

Block	Start Time (μs)
Block 8 & Block 7	t0 + 3.28
Block 6 & Block 5	t0 + 3.28 - 50 * 1
Block 4 & Block 3	t0 + 3.28 - 50 * 2
Block 2 & Block 1	t0 + 3.28 - 50 * 3

■ Laser Firing Time of Each Channel

In each round of firing, the firing sequence is from Channel 1 to Channel 16.

If the start time of Block 6 is t6, the laser firing time of **Channel i** is

$$t6 + [3.024 * (i-1) + 0.28],\ i \in \{1, 2, ..., 16\}.$$

Appendix III PTP Protocol

The Precision Time Protocol (PTP) is used to synchronize clocks across a computer network. It can achieve sub-microsecond clock accuracy.

■ LiDAR Connection When Using PTP

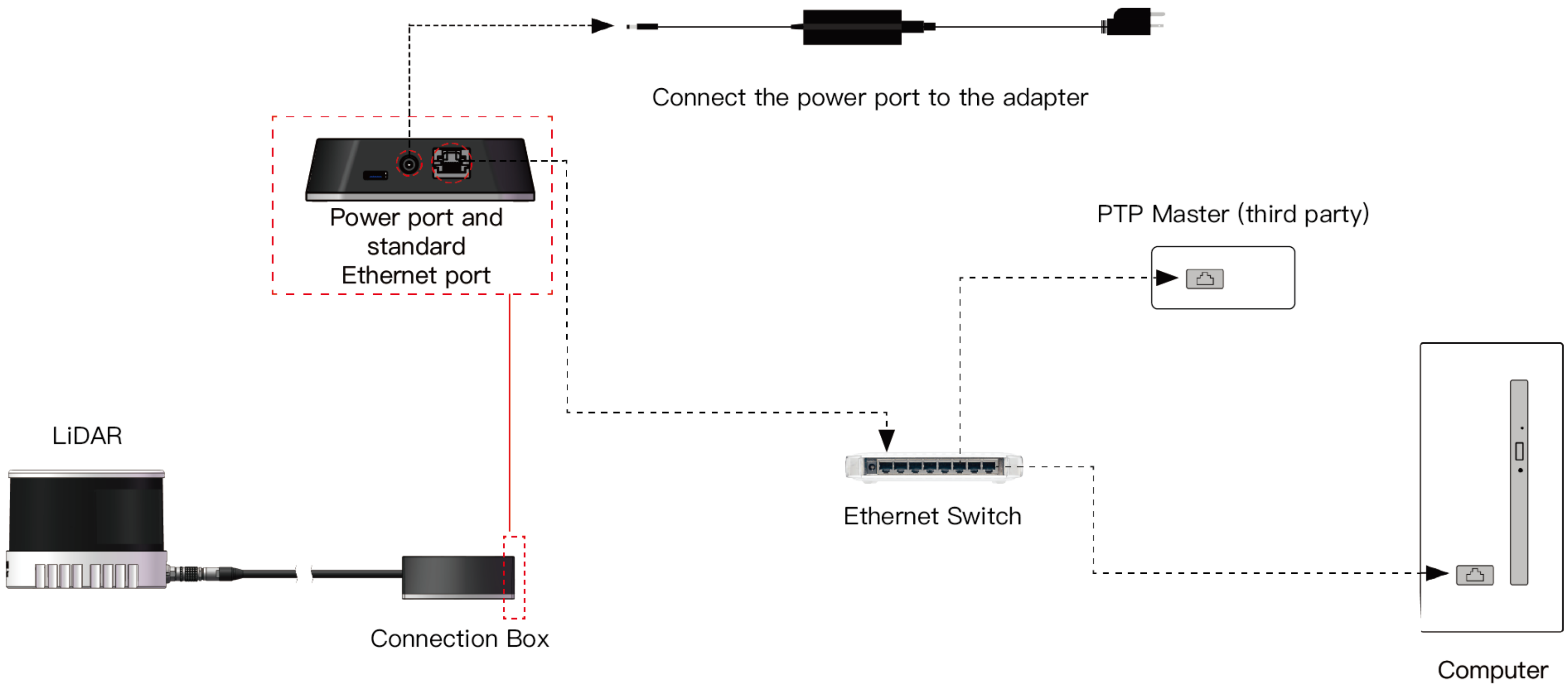


Figure III.1 Connection When Using PTP

■ Absolute Packing Time When Using PTP

To use PTP as the clock source, connect a third-party PTP master device to get the absolute time.

NOTE

- PTP master is a third-party device and is not included with the LiDAR.
- The LiDAR works as a PTP slave device and the PTP protocol is Plug&Play. No additional setup is required.
- When using a PTP clock source, the LiDAR does not output GPS Data Packets.
- The timestamps and Date & Time fields in Point Cloud Data Packets strictly follow the PTP master device. Certain PTP master devices may have a specified offset from the Date & Time output by the LiDAR. Please verify the configuration and calibration of your PTP master device.
- If a PTP clock source is selected but no PTP master device is available, the LiDAR will count the time from an invalid past time. If a PTP clock source is supplied and later stopped, the LiDAR will continue to count the time with an internal clock.

Appendix IV Power Supply Requirements

■ Input Voltage

To ensure that the input voltage at the LiDAR's Lemo connector is 9~36 V DC, please check the specifications of the power source and the voltage drop over cables.

We recommend using 26 AWG cables, which is the thickest wire gauge supported by the LiDAR

- Define the cable length from the power source to the LiDAR's Lemo connector as L (unit: m)
- When using 26 AWG cables, the estimated cable resistance is r = 0.3L (unit: Ω)
- Define the source voltage as U_in (V). The cable voltage drop of the LiDAR operating at 10 Hz under room temperature (23±5°C) can be estimated:

$$U_{drop}(V) = \frac{U_{in} - \sqrt{U_{in}^{2} - 40r}}{2}$$

Users may also estimate the cable voltage drop using the following lookup table.
When cable length exceeds 10 m, source voltage should be at least 24 V.

Estimation of Cable Voltage Drop

Cable Total Length L	Source Voltage $U_{in} = 12$ V	Source Voltage $U_{in} = 24$ V	Source Voltage $U_{in} = 36$ V
1.5 m	0.39 V	0.19 V	0.13 V
2 m	0.52 V	0.25 V	0.17 V
5 m	1.42 V	0.64 V	0.42 V
10 m	3.55 V (LiDAR's input voltage < 9 V)	1.32 V	0.85 V

NOTE When the LiDAR's input voltage approaches 36 V, make sure there is no additional overshoot in the external power system. Even a short period of overvoltage can cause irreversible damage to the LiDAR.

■ Power Consumption

The LiDAR's peak power consumption is below 30 W in all operating conditions.

- After a power-on in an ambient temperature of 0°C or below, power consumption typically remains around 15 W for a period of time.
- When setting the frame rate to 20 Hz, power consumption will also be higher than the typical value in Section 1.4 (Specifications).

In the above or similar conditions, we recommend providing at least 30 W of input power to the LiDAR.

■ Power Up/Down

During a power-up, the voltage requirements are charted in Figure IV.1

- The LiDAR's input voltage should remain under 1 V for more than 50 ms before ramping up
- During the ramp-up, the input voltage should climb to 90% of its designed value in less than 500 ms

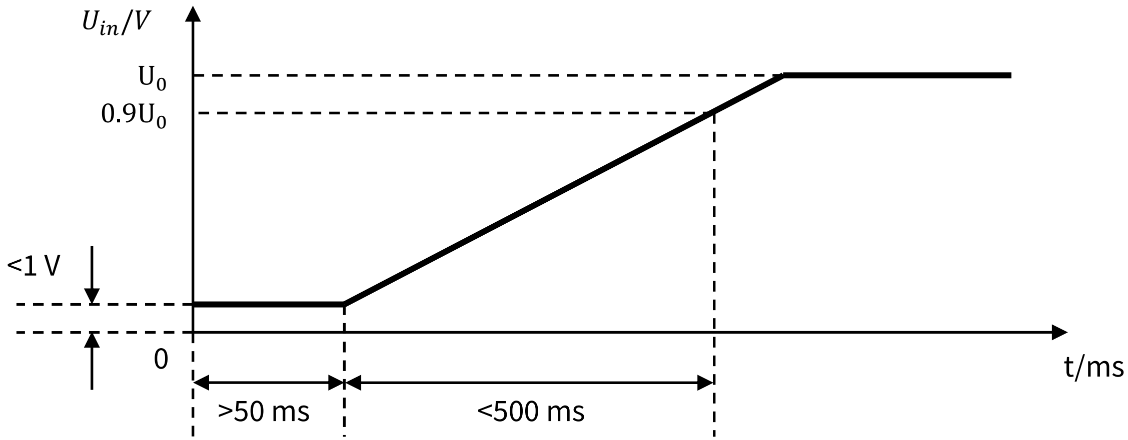


Figure IV.1 Voltage Requirements during a Power-Up

During a power-down, the LiDAR's input voltage, after dropping below 1 V, should remain for more than 50 ms before the next power-up.

Appendix V Nonlinear Reflectivity Mapping

By default, the 1-byte reflectivity data in Point Cloud Data Packets linearly represents target reflectivity from 0 to 255%.

Alternatively, users may choose the Nonlinear Mapping mode, see Chapter 4 (Web Control - Settings).

The nonlinear relationship is detailed below.

Figure V.1 Nonlinear Reflectivity Mapping

Nonlinear Reflectivity Mapping (Continued on the Next Page)

Reflectivity Index (0~255)	Reflectivity (%)	Reflectivity Index (0~255)	Reflectivity (%)	Reflectivity Index (0~255)	Reflectivity (%)	Reflectivity Index (0~255)	Reflectivity (%)
0	0	20	0.67	40	2.69	60	5.9
1	0.01	21	0.75	41	2.81	61	6.1
2	0.02	22	0.81	42	2.94	62	6.3
3	0.03	23	0.87	43	3.07	63	6.5
4	0.04	24	0.95	44	3.21	64	6.7
5	0.05	25	1.05	45	3.36	65	6.9
6	0.08	26	1.15	46	3.5	66	7.1
7	0.11	27	1.25	47	3.64	67	7.3
8	0.13	28	1.35	48	3.79	68	7.5
9	0.15	29	1.45	49	3.93	69	7.7
10	0.19	30	1.55	50	4.08	70	7.9
11	0.23	31	1.65	51	4.25	71	8.12
12	0.26	32	1.75	52	4.42	72	8.37
13	0.29	33	1.85	53	4.58	73	8.62
14	0.34	34	1.95	54	4.75	74	8.87
15	0.39	35	2.06	55	4.92	75	9.1
16	0.44	36	2.19	56	5.1	76	9.3
17	0.5	37	2.31	57	5.3	77	9.5
18	0.56	38	2.44	58	5.5	78	9.7
19	0.61	39	2.56	59	5.7	79	9.9

Nonlinear Reflectivity Mapping (Continued on the Next Page)

Reflectivity Index (0~255)	Reflectivity (%)	Reflectivity Index (0~255)	Reflectivity (%)	Reflectivity Index (0~255)	Reflectivity (%)	Reflectivity Index (0~255)	Reflectivity (%)
80	10.17	100	15.87	120	22.83	140	31.17
81	10.5	101	16.17	121	23.25	141	31.5
82	10.83	102	16.5	122	23.75	142	31.83
83	11.12	103	16.83	123	24.17	143	32.25
84	11.37	104	17.17	124	24.5	144	32.75
85	11.62	105	17.5	125	24.83	145	33.25
86	11.87	106	17.83	126	25.25	146	33.75
87	12.12	107	18.17	127	25.75	147	34.25
88	12.37	108	18.5	128	26.17	148	34.75
89	12.62	109	18.83	129	26.5	149	35.25
90	12.87	110	19.17	130	26.83	150	35.75
91	13.17	111	19.5	131	27.25	151	36.25
92	13.5	112	19.83	132	27.75	152	36.75
93	13.83	113	20.25	133	28.17	153	37.25
94	14.17	114	20.75	134	28.5	154	37.75
95	14.5	115	21.17	135	28.83	155	38.25
96	14.83	116	21.5	136	29.25	156	38.75
97	15.12	117	21.83	137	29.75	157	39.17
98	15.37	118	22.17	138	30.25	158	39.5
99	15.62	119	22.5	139	30.75	159	39.83

Nonlinear Reflectivity Mapping (Continued on the Next Page)

Reflectivity Index (0~255)	Reflectivity (%)	Reflectivity Index (0~255)	Reflectivity (%)	Reflectivity Index (0~255)	Reflectivity (%)	Reflectivity Index (0~255)	Reflectivity (%)
160	40.5	180	51.25	200	63.25	220	76.5
161	41.25	181	51.75	201	63.75	221	77.25
162	41.75	182	52.25	202	64.5	222	77.75
163	42.25	183	52.75	203	65.25	223	78.5
164	42.75	184	53.5	204	65.75	224	79.25
165	43.25	185	54.25	205	66.25	225	79.75
166	43.75	186	54.75	206	66.75	226	80.5
167	44.25	187	55.25	207	67.5	227	81.25
168	44.75	188	55.75	208	68.25	228	81.75
169	45.25	189	56.5	209	68.75	229	82.5
170	45.75	190	57.25	210	69.5	230	83.5
171	46.25	191	57.75	211	70.25	231	84.25
172	46.75	192	58.25	212	70.75	232	84.75
173	47.25	193	58.75	213	71.5	233	85.5
174	47.75	194	59.5	214	72.25	234	86.5
175	48.25	195	60.25	215	72.75	235	87.25
176	48.75	196	60.75	216	73.5	236	87.75
177	49.5	197	61.25	217	74.25	237	88.5
178	50.25	198	61.75	218	74.75	238	89.25
179	50.75	199	62.5	219	75.5	239	89.75

Nonlinear Reflectivity Mapping (Continued)

Reflectivity Index (0~255)	Reflectivity (%)
240	90.5
241	91.5
242	92.5
243	93.25
244	93.75
245	94.5
246	95.5
247	96.25
248	96.75
249	97.5
250	98.5
251	99.5
252	132
253	196
254	242

Appendix VI Certification Info

■ FCC Declaration

FCC ID: 2ASO2PANDARXT

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:
(1) this device may not cause harmful interference, and
(2) this device must accept any interference received, including interference that may cause undesired operation.

Caution
The user is cautioned that changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

Note
This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:
—Reorient or relocate the receiving antenna.
—Increase the separation between the equipment and receiver.
—Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
—Consult the dealer or an experienced radio/TV technician for help.

■ IC Statement

This device complies with Industry Canada licence-exempt RSS standard(s).

Operation is subject to the following two conditions:
(1) this device may not cause interference, and
(2) this device must accept any interference, including interference that may cause undesired operation of the device.

Le présent appareil est conforme aux CNR d'Industrie Canada applicables aux appareils radio exempts de licence.

L'exploitation est autorisée aux deux conditions suivantes:
(1) l'appareil ne doit pas produire de brouillage, et
(2) l'utilisateur de l'appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement.

■ NOTE

This product is only suitable for industrial use.

Appendix VII Support and Contact

■ Technical Support

If your question is not addressed in this manual, please contact us at:

service@hesaitech.com
www.hesaitech.com
https://github.com/HesaiTechnology

NOTE Please leave your questions under the corresponding GitHub projects.

■ Legal Notice

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HESAI Wechat